Irregular Repetition Slotted ALOHA with Multi-Antenna Reception: Slot-Level Analysis and Fundamental Limits

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***Abstract*—Multi-antenna reception with successive interference cancellation (SIC) can substantially improve collision resolution in random access systems. However, analytically characterizing decoding error probabilities within a slot over fading channels is challenging because SIC couples decoding events across antennas. To facilitate analysis, we consider selection-combining-based SIC (SC-SIC), which decodes packets from selected antenna branches without coherent combining. Under Rayleigh block-fading channels, we apply the inclusion–exclusion principle to derive exact closed-form expressions for the average decoding error probability $\bar{\epsilon}_{m,L}$ in an m-packet collision for $m \leq 2$ and a closed-form approximation for $m \geq 3$, both applicable to any finite number of receive antennas L and any spectral efficiency $R > 0$. We characterize error floors at high signal-to-noise ratios and identify collision-order-dependent spectral-efficiency conditions under which they vanish. These probabilities encapsulate within-slot decoding for irregular repetition slotted ALOHA (IR-SA). With the aid of density evolution (DE), we characterize fundamental limits through belief-propagation (BP) thresholds. We also characterize the maximum a posteriori (MAP) thresholds and derive a converse bound. Numerical results show that the MAP thresholds lie close to the corresponding converse bounds, demonstrating threshold saturation under multi-antenna reception.**

***Index Terms*—Average decoding error probability, high-SNR error floor, irregular repetition slotted ALOHA, multi-antenna reception, successive interference cancellation.**

I. Introduction

Massive machine-type communication (mMTC) is expected to be a key enabler of future internet-to-everything networks [1]. In mMTC systems, a large number of devices are typically connected to each access point (AP). At every time-slot, only a small fraction of them are active in a sporadic manner [2]. In this case, random access (RA) is more suitable than coordinated access schemes that require prior scheduling or dedicated resource allocation, as it can support uplink mMTC transmission with reduced access latency and signaling overhead. Among various RA schemes, those evolving from slotted ALOHA (SA) [3] have garnered significant attention, as they allow devices to transmit in an 'arrive-and-go' manner. Following this research line, a key challenge is how to analytically derive the decoding error probability of colliding packets under multi-antenna reception over fading channels.

A. Problem Description and Contributions

In SA systems, we consider an m-packet collision at an AP with L receive antennas. Each antenna observes a superposition of the same m packets over Rayleigh block-fading channels. Owing to independent fading across antennas, the received signal-to-noise ratios (SNRs) of the colliding packets vary across antennas.

A key objective is to derive analytical expressions for the average decoding error probability $\bar{\epsilon}_{m,L}$ in an m-packet collision under L-antenna reception over Rayleigh block-fading channels. Here, $\bar{\epsilon}_{m,L}$ is the probability that a randomly selected reference packet remains undecoded after the intra-slot decoding process, averaged over the fading realizations. We refer to this characterization as a *slot-level analysis*.

For multi-antenna reception, Srivatsa *et al.* [4, Theorem 3] characterized the average decoding probability θ_m of a reference packet in an m-packet collision, under maximal-ratio-combining-based successive interference cancellation (MRC-SIC) over Rayleigh block-fading channels with perfect channel state information (CSI). Here, the dependence of θ_m on L is implicit in their notation, and θ_m is analogous to $1 - \bar{\epsilon}_{m,L}$ in our work. For collided slots, an exact analysis for finite L is challenging because the post-combining signal-to-interference-plus-noise ratios (SINRs) depend jointly on random channel norms and inner products between different users' channel vectors, and are coupled across packets and successive SIC steps. An exact expression was derived for a singleton slot ($m = 1$) for any finite L, whereas collided slots ($m \geq 2$) were treated using closed-form large-antenna approximations.

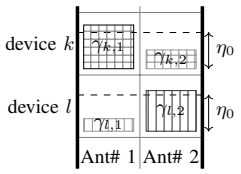
In this paper, we focus on the slot-level analysis of selection-combining-based SIC (SC-SIC), in which packet decoding uses a selected antenna branch without coherent signal combining. At each SIC step, the receiver selects the packet–antenna pair with the highest instantaneous SINR among all remaining pairs. If this SINR exceeds the decoding threshold $\eta_0 = 2^R - 1$, where R denotes the spectral efficiency, the packet is decoded and canceled from all antenna observations within the slot; otherwise, decoding within that slot stops. Figure 1 illustrates this SC-SIC process, including intra- and

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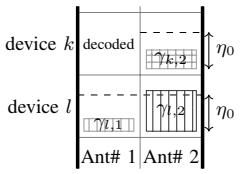
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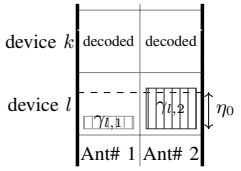
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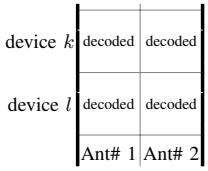
(a) Two packets are received at the two antennas.



(b) At Ant# 1, the packet associated with $\gamma_{k,1}$ is decoded using intra-antenna SIC.



(c) Since $\gamma_{l,1} < \eta_0$, Ant# 1 does not perform decoding. The signal from device k is canceled at Ant# 2 using inter-antenna SIC.



(d) Since $\gamma_{l,2} > \eta_0$, the packet of device l is decoded, and its signal at Ant# 1 is canceled using inter-antenna SIC.

Fig. 1: An example of intra- and inter-antenna SIC decoding within a two-packet slot. The AP is equipped with $L = 2$ antennas.

inter-antenna interference cancellation, within a two-packet slot with $L = 2$ receive antennas.

For single-antenna reception ($L = 1$), SC-SIC reduces to conventional intra-slot SIC. In this case, Clazzer *et al.* [5] derived an exact closed-form expression for $\bar{\epsilon}_{m,1}$ over Rayleigh block-fading channels. Their analysis considers threshold-based intra-slot SIC and is applicable to any finite m for $R \geq 1$.

Having said this, an exact characterization of $\bar{\epsilon}_{m,L}$ under SC-SIC for arbitrary finite m and L remains an open problem.

Under SC-SIC, for an m-packet collision at an L-antenna AP, the decoding outcome depends jointly on the mL branch-wise fading realizations. Moreover, once a packet is decoded at one antenna, canceling its interference at the others changes their instantaneous SINRs and may trigger further decoding. Thus, decoding events across antennas are coupled through iterative SIC, and the number of decoding-event patterns grows rapidly with m and L, making $\bar{\epsilon}_{m,L}$ difficult to characterize.

To address this challenge, we first characterize the SIC-feasible SNR regions for a two-packet collision separately for $R \geq 1$ and $0 < R < 1$ and derive the associated closed-form probabilities. Using these probabilities together with the inclusion–exclusion principle across the receive antennas, we derive exact closed-form expressions for $\bar{\epsilon}_{m,L}$ for collision sizes $m \in \{1, 2\}$, any finite L, and $R > 0$. For $m \geq 3$, we develop a closed-form approximation by combining the exact first-step decoding probability for an arbitrary collision size with the exact two-packet result. The approximation is likewise applicable to any finite L and $R > 0$.

For fixed finite L, the $R < 1$ regime is of particular importance because the *collision-order-dependent spectral-efficiency condition* for a vanishing high-SNR error floor imposes an upper bound on R that decreases with the collision order and is strictly below 1 for $m \geq 3$.

The resulting slot-level decoding probabilities characterize SC-SIC decoding within a single slot and are directly applicable to conventional SA with L-antenna reception. Here, conventional SA refers to SA without packet repetition or inter-slot SIC, while allowing intra-slot SIC for collision resolution. Averaging $\bar{\epsilon}_{m,L}$ over the distribution of the number of packets interfering with a reference packet yields the packet loss rate (PLR), from which the throughput is obtained (see **Remark 2**).

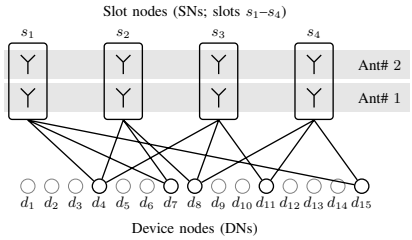


Fig. 2: Bipartite graph representation of multi-antenna IR-SA with $L = 2$, $M = 4$, $N_{\mathrm{T}} = 15$, $N_{\mathrm{a}} = 5$, and $\pi = N_{\mathrm{a}}/N_{\mathrm{T}} = 1/3$.

We further apply the derived decoding probabilities to irregular repetition slotted ALOHA (IR-SA), which employs packet repetition and inter-slot SIC to exploit temporal diversity [5], [6]. Figure 2 illustrates an example of the bipartite graph representation of multi-antenna IR-SA, with device nodes connected to slot nodes by edges representing packet replicas. Following the density-evolution (DE) analyses that incorporate slot-level decoding probabilities [4], [5], we analyze multi-antenna IR-SA using a DE recursion in which the effect of L-antenna SC-SIC processing within each slot node is encapsulated by the reference-packet decoding success probability $1-\bar{\epsilon}_{m,L}$. The resulting DE framework characterizes the belief propagation (BP) thresholds of multi-antenna IR-SA.

Next, we incorporate the SC-SIC decoding probabilities into the DE analysis of multi-antenna IR-SA. Specifically, L-antenna processing within each slot node (SN) is encapsulated by the decoding success probability $1-\bar{\epsilon}_{m,L}$, while the graph-level DE recursion retains the same form as in the single-antenna case. This enables the BP threshold of IR-SA with SC-SIC to be characterized for any finite L.

We futher characterize fundamental performance limits of multi-antenna IR-SA under SC-SIC through converse-bound and maximum a posteriori (MAP)-threshold analyses. Building on our previous single-antenna study [7], we extend the extrinsic information transfer (EXIT)-area and trial-entropy frameworks from $L = 1$ to any finite L using the L-antenna SN decoding function $\chi_L(\cdot)$, determined by $1 - \bar{\epsilon}_{m,L}$.

Numerical results validate the proposed slot-level and DE analyses and show that the MAP thresholds lie close to the converse bounds while spatial coupling drives the BP thresholds toward the MAP thresholds.

B. Related Works

For conventional SA with single-antenna reception, intra-slot SIC enhances multi-packet reception (MPR) over fading channels. In [8], the packet decoding probability for an arbitrary collision size m was analytically characterized under received-power-ordered intra-slot SIC over Rayleigh fading channels and used to derive the maximum sum rate. In [9], closed-form expressions for the average throughput of non-orthogonal multiple access (NOMA)-based SA that resolves

TABLE I
Comparison of our work with selected studies on slot-level reception over fading channels and applications to conventional SA and IR-SA.

Work	Receiver architecture	Channel model	Intra-slot decoding	Slot-level decoding characterization	Asymptotic analysis for IR-SA		
Conventional SA					*Not applicable to conventional SA*		
[8]	Single-antenna AP	Rayleigh fading	Power-ordered intra-slot SIC	Exact for any m			
[9]	Single-antenna AP	Nakagami fading	NOMA-based intra-slot SIC	Exact for $m \leq 2$			
[10], [11]	Multi-antenna AP	Rayleigh fading [10], Rayleigh fading with log-normal shadowing [11]	Capture effect without SIC	Conditional throughput for finite m; large-m limit			
IR-SA					DE / BP threshold	MAP threshold	Converse bound
[5], [7]	Single-antenna AP	Rayleigh fading	Threshold-based SIC (perfect CSI)	Exact for any m, $R \geq 1$	✓	✓ [7]	✓ [7]
[12]	Multiple APs	Path loss, shadowing, and small-scale fading	Threshold-based cooperative SIC	Monte Carlo simulation	✓	–	–
[4]	Multi-antenna AP	Rayleigh fading	MRC-SIC (perfect CSI)	Exact for $m = 1$ and any finite L; large-L approximation for $m \geq 2$	✓	–	–
This work	Multi-antenna AP	Rayleigh fading	SC-SIC (perfect CSI)	For any finite L and $R > 0$: exact for $m \in \{1, 2\}$; approximation for $m \geq 3$	✓	✓	✓

Note: Boldface in the Work column highlights the most closely related studies and the present work. The slot-level decoding characterization derived in this paper also enables the performance analysis of conventional SA with L-antenna SC-SIC reception (see **Remark 2**).

two-packet collisions via intra-slot SIC over Nakagami-m fading channels were derived. Cross-slot SIC has also been investigated for packet recovery in random NOMA [13], [14].

Spatial-diversity reception has also been investigated in conventional SA. The capture effect at a multi-antenna AP was analyzed in [10], [11]. In these studies, each antenna independently decodes a packet whose signal-to-interference ratio (SIR) exceeds a predetermined threshold, without interference cancellation. In [11], the conditional throughput for a given collision size and its large-collision limit were analyzed.

Under the collision-channel model, graph-based analysis and code design provide a framework for IR-SA and related coded RA schemes that exploit temporal diversity across slots [6], [15], building on iterative decoding principles from coding theory [16]. This connection has also motivated spatially coupled RA constructions [17].

Spatial-diversity reception using multiple receivers has been investigated for collision resolution, with applications to IR-SA and related graph-based RA schemes through joint spatio-temporal SIC [18]–[20]. These analyses, however, mainly employ packet-erasure models [18], [19] or abstract receiver models [20] and do not analytically characterize the joint behavior of instantaneous SINRs across fading branches.

Under an explicit fading model, [12] studied cooperative reception with intra-slot, inter-slot, and inter-AP SIC for NOMA-based IR-SA. In their analysis, the probability that a packet in a collision is successfully decoded within a slot, analogous to $1 - \bar{\epsilon}_{m,L}$ in our framework, is evaluated numerically using Monte Carlo simulations rather than derived in closed form. The resulting probabilities are then used in a DE-based performance analysis of cooperative IR-SA.

Performance bounds for single-antenna graph-based RA have been studied over collision [17], [21], on-off fading [22], Rayleigh block-fading [7], [23], and Gaussian channels [24]. EXIT-area arguments and trial-entropy analysis provide tools for characterizing these limits [7], [17], [25].

Parallel decoding of single-antenna IR-SA has also been studied, including asymptotic throughput analysis for two-replica IR-SA over additive white Gaussian noise (AWGN) channels [26] and simulation-based evaluation of partially parallel decoding over AWGN and Rayleigh fading channels, highlighting the tradeoffs among decoding latency, complexity, and throughput [27]. In addition, parallel decoding of IR-SA with single-antenna reception has been studied. Chiang *et al.* [26] derived asymptotic throughput expressions for the two-replica case over additive white Gaussian noise (AWGN) channels. Shieh *et al.* [27] proposed partially parallel decoding over AWGN and Rayleigh fading channels, evaluating the tradeoffs among decoding latency, complexity, and throughput through simulations.

Table I summarizes selected studies on slot-level reception over fading channels and their applications to conventional SA and IR-SA.

C. Paper Organization

The remainder is organized as follows. Section II describes the intra-slot SC-SIC procedure and analyzes decoding events for a two-packet collision at a single antenna. Section III derives exact and approximate expressions for the average decoding error probability $\bar{\epsilon}_{m,L}$. Section IV presents the system model for IR-SA with multi-antenna reception over fading channels. Section V provides the DE equations. Using these equations, Section VI derives the converse bound and the MAP threshold. Finally, numerical and simulation results are discussed in Sec. VII, followed by the conclusion in Sec. VIII.

II. Multi-Antenna SC-SIC Decoding in a Slot

In this section, we describe the SC-SIC decoding procedure for an m-packet collision within a single slot at an AP equipped with L receive antennas. The corresponding average decoding error probability $\bar{\epsilon}_{m,L}$ is derived in Sec. III.

A. Intra-Slot SIC

Let $\mathcal{L} = \{1, 2, \ldots, L\}$ denote the set of receive antenna indices. Assuming that m packets collide, the resultant L superimposed signals (one per antenna) within the slot are

$$\boldsymbol{y}_i = \textstyle\sum_{k \in \mathcal{K}} h_{k,i} \boldsymbol{x}_k + \boldsymbol{z}_i, \quad i = 1, 2, \ldots, L, \tag{1}$$

where $\mathcal{K}$ ($|\mathcal{K}| = m$) is the set of active devices. Here, m devices are active and have accessed the slot. $\boldsymbol{x}_k$ represents device k's codeword (or packet) of code $\mathcal{C}$ with a spectral efficiency R. $\boldsymbol{z}_i \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 I)$ is an additive circularly symmetric complex Gaussian noise with mean $\boldsymbol{0}$ and diagonal covariance matrix $\sigma^2 I$ with noise power σ^2 and identity matrix I. All devices transmit at a common power P. The channel coefficient $h_{k,i}$, which characterizes the channel from active device k to Ant# i of the AP, remains constant within a slot and varies independently across slots. The antennas at

the AP are spaced sufficiently far apart to ensure independent fading values. For Rayleigh block-fading channels, the received SNR $\gamma_{k,i} \triangleq P|h_{k,i}|^2/\sigma^2$ is exponentially distributed with $p_\gamma(x)=1/\bar{\gamma}\cdot e^{-\frac{x}{\bar{\gamma}}},\ x>0$, where $\bar{\gamma}$ is the average received SNR.

Decoding at the AP within a slot involves an iterative process alternating between intra-antenna and inter-antenna SIC. As in [5], [7], [9], we assume that the AP has perfect knowledge of the CSI and the number of colliding packets. We consider SC-SIC, in which the receiver selects the packet–antenna pair with the largest instantaneous SINR among all remaining pairs for decoding.

1) Intra-Antenna SIC: For an m-packet collision at Ant# i, assuming $\gamma_{1,i} > \gamma_{2,i} > \cdots > \gamma_{m,i}$, the SINR of the packet from device k is given by

$$\eta_{k,i} \triangleq \frac{P|h_{k,i}|^2}{\sigma^2 + \sum_{u=k+1}^{m} P|h_{u,i}|^2} = \frac{\gamma_{k,i}}{1 + \sum_{u=k+1}^{m} \gamma_{u,i}}. \quad (2)$$

Under threshold-based decoding, packet $\boldsymbol{x}_k$ is successfully decoded if and only if $\eta_{k,i} > \eta_0$, where η_0 satisfies $R = \log_2(1+\eta_0)$ [28]. Upon successful decoding, the packet is removed from the received signal $\boldsymbol{y}_i$. Subsequent decoding proceeds with the interference-removed signal. This iterative process is termed *intra-antenna* SIC.

An undecoded packet $\boldsymbol{x}_k$ with SNR above η_0, i.e., $\gamma_{k,i} > \eta_0$, is termed a *potentially decodable (PD)* packet [13], because it may become decodable after one or more interfering packets $\boldsymbol{x}_{j\neq k}$ are removed from the signal $\boldsymbol{y}_i$.

2) Inter-Antenna SIC: The intra-antenna SIC process enables the decoding of packets. These decoded packets are then canceled from the signals received at the other antennas. This procedure is referred to as *inter-antenna* SIC.

The above two processes are repeated until no further packet can be decoded at any antenna. This iterative decoding procedure, illustrated in Fig. 1, is referred to as *intra-slot* SIC.

B. Probabilities in a Two-Packet Collision at a Single Antenna

In Sec. III, the average decoding error probability $\bar{\epsilon}_{m,L}$ is formulated in terms of the probabilities of the decoding events at an individual antenna, where the events of a two-packet collision play a key role. In this subsection, we classify the decoding events of a two-packet collision at a single antenna and derive closed-form expressions for the probabilities associated with the different SNR regions.

We consider the signal received at Ant# i in a slot where two packets collide, which follows from (1) with $m=2$. Since the following analysis concerns a single antenna, we omit the antenna index i and write $\boldsymbol{y} = \sum_{k\in\mathcal{K}} h_k \boldsymbol{x}_k + \boldsymbol{z}$, where $\mathcal{K}=\{k,l\}$ denotes the set of active devices.

Before deriving the probability of two packets being received in distinct SNR regions, we define $\Gamma = \{(\gamma_k,\gamma_l)|\gamma_k\in\Gamma_k,\gamma_l\in\Gamma_l\}$ as a probability region, with $\Gamma_k, \Gamma_l \subseteq \mathcal{R}^+$ being subsets of the non-negative real numbers $\mathcal{R}^+$.

For the two independent random variables (RVs) γ_k and γ_l, the probability that (γ_k,γ_l) lies in region Γ is given by

$$P_\Gamma = \Pr\{(\gamma_k,\gamma_l)|\gamma_k\in\Gamma_k,\gamma_l\in\Gamma_l\} = \int_{\Gamma_l} p(\gamma_l) \int_{\Gamma_k} p(\gamma_k) \mathrm{d}\gamma_k \mathrm{d}\gamma_l.$$

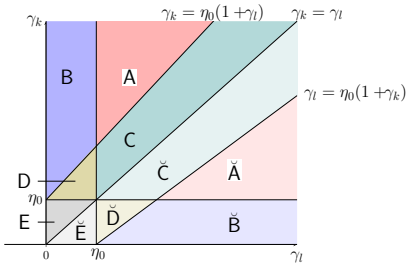


Fig. 3: The SIC feasible region in a two-packet collision ($m=2$) between devices k and l under $R \geq 1$.

TABLE II
Events at a single antenna, where P_k and P_l are packets sent by devices k and l, respectively.

region	event
A (Ă)	P_k (P_l) is decoded first and then P_l (P_k) is decoded.
B (B̆)	P_k (P_l) is decoded, while P_l (P_k) is undecoded and not PD.
C, Č	P_k and P_l are PD.
D (Ď)	P_k (P_l) is undecoded and PD; P_l (P_k) is undecoded and not PD.
E, Ě	P_k and P_l are undecoded and not PD.

The probability expressions are divided into two cases: $R \geq 1$ and $0 < R < 1$, as detailed below.

1) Closed-Form Expressions for the Probabilities ($R \geq 1$): In the case of $R \geq 1$, Fig. 3 illustrates the SNR regions A to E ($\gamma_k > \gamma_l$) along with their symmetric counterparts Ă to Ě ($\gamma_k < \gamma_l$). The events listed in Table II are defined.

For $\gamma_k > \gamma_l$, the probabilities for regions A to E are:

$$P_{\mathsf{A}} = \Pr\{\mathsf{A}:(\gamma_k,\gamma_l)|\gamma_k > \eta_0(1+\gamma_l), \gamma_l > \eta_0\} \quad (3a)$$
$$P_{\mathsf{B}} = \Pr\{\mathsf{B}:(\gamma_k,\gamma_l)|\gamma_k > \eta_0(1+\gamma_l), \gamma_l \leq \eta_0\} \quad (3b)$$
$$P_{\mathsf{C}} = \Pr\{\mathsf{C}:(\gamma_k,\gamma_l)|\gamma_l < \gamma_k \leq \eta_0(1+\gamma_l), \gamma_l > \eta_0\} \quad (3c)$$
$$P_{\mathsf{D}} = \Pr\{\mathsf{D}:(\gamma_k,\gamma_l)|\eta_0 < \gamma_k \leq \eta_0(1+\gamma_l), \gamma_l \leq \eta_0\} \quad (3d)$$
$$P_{\mathsf{E}} = \Pr\{\mathsf{E}:(\gamma_k,\gamma_l)|\gamma_l < \gamma_k \leq \eta_0, \gamma_l \leq \eta_0\}. \quad (3e)$$

For $\gamma_k < \gamma_l$, the probabilities for regions Ă to Ě are defined analogously. In Appendix I-A, we derive the closed-form expressions for probabilities P_{A} through P_{E} while excluding $P_{\check{\mathsf{A}}}$ and $P_{\check{\mathsf{E}}}$ due to symmetry.

2) Closed-Form Expressions for the Probabilities ($0 < R < 1$): In the case of $R < 1$, the decoding threshold η_0 is less than 1, i.e., $\eta_0 < 1$. This implies that the two lines $\gamma_k = \gamma_l$ and $\gamma_k = \eta_0(1+\gamma_l)$ (or $\gamma_l = \eta_0(1+\gamma_k)$) intersect at a point denoted by $(\frac{\eta_0}{1-\eta_0}, \frac{\eta_0}{1-\eta_0})$ (see Fig. 4), where $\frac{\eta_0}{1-\eta_0}$ is the unique solution to equation $\gamma_k = \eta_0(1+\gamma_k)$ (or $\gamma_l = \eta_0(1+\gamma_l)$).

This implies that the probability that both packets are decodable over the entire γ_k–γ_l plane is given by

$$\Pr\{\mathsf{A}'\cup\check{\mathsf{A}}'\cup\mathsf{F}'\cup\check{\mathsf{F}}'\} = (P_{\mathsf{A}'\cup\mathsf{F}'\cup\check{\mathsf{F}}'} - P_{\check{\mathsf{F}}'}) + (P_{\check{\mathsf{A}}'\cup\mathsf{F}'\cup\check{\mathsf{F}}'} - P_{\mathsf{F}'}).$$

Note that the ranges of the double integral used to compute $P_{\mathsf{A}'\cup\mathsf{F}'\cup\check{\mathsf{F}}'}$ are the same as those in (3a).

To treat both ranges of R in a unified manner, as in the case of $R \geq 1$, we give the probabilities for the case $\gamma_k > \gamma_l$:

$$\begin{aligned} P_{(\mathsf{A}'\cup\mathsf{F}')} &= P_{\mathsf{A}'\cup\mathsf{F}'\cup\check{\mathsf{F}}'} - P_{\check{\mathsf{F}}'} \\ &= \Pr\{(\gamma_k,\gamma_l)|\gamma_k > \eta_0(1+\gamma_l), \gamma_l > \eta_0\} \\ &\quad - \Pr\left\{(\gamma_k,\gamma_l)|\eta_0(1+\gamma_l) < \gamma_k < \gamma_l, \gamma_l > \tfrac{\eta_0}{1-\eta_0}\right\}, \end{aligned}$$

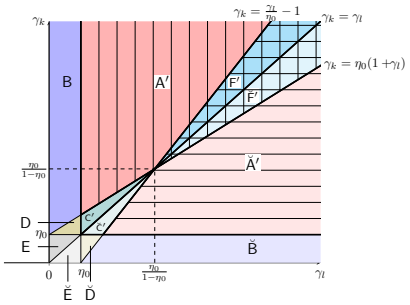


Fig. 4: The SIC feasible region in a two-packet collision between devices k and l in a slot under $0 < R < 1$. Vertical lines illustrate the region $(\mathsf{A}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}')$, while horizontal lines represent the region $(\breve{\mathsf{A}}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}')$.

$$P_{\mathsf{C}'} = \Pr\Big\{(\gamma_k, \gamma_l) | \gamma_l < \gamma_k < \eta_0(1+\gamma_l), \eta_0 < \gamma_l < \tfrac{\eta_0}{1-\eta_0}\Big\}.$$

The closed-form expressions for $P_{(\mathsf{A}' \cup \mathsf{F}')}$ and $P_{\mathsf{C}'}$ in Appendix I-B. Note that the closed-form expressions for P_{B}, P_{D}, and P_{E} have the same forms for $R \geq 1$ and $0 < R < 1$. The probabilities for regions $(\breve{\mathsf{A}}' \cup \breve{\mathsf{F}}')$ and $\breve{\mathsf{C}}'$ $(\gamma_k < \gamma_l)$ can be derived in a similar manner, and are omitted for brevity.

Since $P_{(\mathsf{A}' \cup \mathsf{F}')}$ and P_{A} represent the probability that two packets are successfully decoded, we adopt the unified notation P_{A} and extend it to the regime $0 < R < 1$. In this range, P_{A} follows the expression originally derived for $P_{(\mathsf{A}' \cup \mathsf{F}')}$, whereas for $R \geq 1$, it follows the expression previously denoted by P_{A}. Likewise, the probability P_{C} is extended to $0 < R < 1$ by employing its counterpart $P_{\mathsf{C}'}$ defined for this regime.

Although the above derivations are separated into $R \geq 1$ and $0 < R < 1$, the final expressions can be unified for $R > 0$, with only some intermediate formulas retaining the two-case form. This allows $\bar{\epsilon}_{m,L}$ to be derived for $R > 0$ without distinguishing between the two cases. While many existing slot-level analyses focus solely on the $R \geq 1$ regime (e.g., [5], [7], [13], [14], [23]), practical systems often operate with $0 < R < 1$ because of multi-device interference and the need for low-rate channel coding with low-order modulation [29].

III. Average Decoding Error Probability in a Slot

This section considers a slot in which m packets collide at an AP equipped with L receive antennas, as in (1). Among the m colliding packets, one packet is selected uniformly at random and designated as the reference packet. Owing to the statistical symmetry among the packets, the average decoding error probability does not depend on which packet is selected as the reference. Under intra-slot SIC, we derive exact closed-form expressions for $\bar{\epsilon}_{m,L}$ for $m = 1, 2$ and, based on these results, a closed-form approximation for $m \geq 3$.

In threshold-based decoding, a single decoding step entails the recovery of a packet at any of the L receive antennas, followed by its interference cancellation across all antennas. Consequently, resolving an m-packet collision may require up to m successive steps. Let $D(m, r, L)$ denote the probability that the reference packet is decoded exactly at the r-th step. The average decoding probability is then given by $D(m, L) = \sum_{r=1}^{m} D(m, r, L)$. Thus, the average decoding error probability for the reference packet, given $m-1$ interferers within a slot and L receive antennas, is expressed as

$$\bar{\epsilon}_{m,L} = 1 - D(m, L) = 1 - \textstyle\sum_{r=1}^{m} D(m, r, L). \tag{4}$$

To derive $\bar{\epsilon}_{m,L}$ while accounting for the interactions among decoding events across multiple antennas, we rely on the following fundamental facts:

(i) RVs $\gamma_{k,i}$ are independent and identically distributed.
(ii) The inclusion-exclusion principle for computing the probability of the union of L sets $A_1, \ldots, A_L$ [30]:

$$\Pr\Big(\bigcup_{\ell=1}^{L} A_\ell\Big) = \sum_{\emptyset \neq \mathcal{S} \subseteq \{1,\ldots,L\}} (-1)^{|\mathcal{S}|+1} \Pr\Big(\bigcap_{\ell \in \mathcal{S}} A_\ell\Big). \tag{5}$$

A. Exact Closed-Form Decoding Probabilities for $m \leq 2$

1) Single-Packet Slot ($m = 1$): The probability that the packet of device k, which is the only packet received in the slot, is decoded by at least one of the antennas is given by

$$D(1,1,L) = 1 - \prod_{i=1}^{L} (1 - \Pr\{\gamma_{k,i} > \eta_0\}) = 1 - (1 - e^{-\frac{\eta_0}{\bar{\gamma}}})^L. \tag{6}$$

2) Two-Packet Slot ($m = 2$):

a) $D(2,1,L)$: We derive the probability that the reference packet is successfully decoded in the first threshold-based decoding step. This probability is expressed as

$$D(2,1,L) = \frac{1}{2}\Pr\{\Upsilon\} = \frac{1}{2}\big(1 - \{1 - 2(P_{\mathsf{A}} + P_{\mathsf{B}})\}^L\big). \tag{7}$$

$\Pr\{\Upsilon\}$ denotes the probability that either P_k or P_l is decodable at any of the L receive antennas in the first step. Note that the reference packet is either P_k or P_l with probability $1/2$. For simplicity, we first consider the case where either P_k or P_l is decodable at a specific antenna $i \in \mathcal{L}$ in the first step. The probability of this event is given by $\Pr\{\Upsilon_i\} = \Pr\{\mathsf{A}_i\} + \Pr\{\mathsf{B}_i\} + \Pr\{\breve{\mathsf{A}}_i\} + \Pr\{\breve{\mathsf{B}}_i\} = 2(P_{\mathsf{A}} + P_{\mathsf{B}})$. Thus, the probability that one of the two packets is first decoded at any antenna in $\mathcal{L}$ is $\Pr\{\Upsilon\} = \Pr\{\bigcup_{i \in \mathcal{L}} \Upsilon_i\}$. Using the inclusion-exclusion principle (5) and the binomial theorem, we obtain

$$\begin{aligned}
\Pr\{\Upsilon\} = \Pr\{\bigcup_{i \in \mathcal{L}} \Upsilon_i\} &= \sum_{\emptyset \neq \mathcal{S} \subseteq \mathcal{L}} (-1)^{|\mathcal{S}|+1} \Pr\Big(\bigcap_{i \in \mathcal{S}} \Upsilon_i\Big) \\
&= \sum_{\ell=1}^{L} \sum_{\substack{\mathcal{S} \subseteq \mathcal{L} \\ |\mathcal{S}| = \ell}} (-1)^{\ell+1} \Pr\Big(\bigcap_{i \in \mathcal{S}} \Upsilon_i\Big) \\
&= \sum_{\ell=1}^{L} (-1)^{\ell+1} \binom{L}{\ell} \{2(P_{\mathsf{A}} + P_{\mathsf{B}})\}^\ell \\
&= 1 - \{1 - 2(P_{\mathsf{A}} + P_{\mathsf{B}})\}^L.
\end{aligned} \tag{8}$$

Here, the expression is transformed by noting that, for a fixed $\ell > 0$, there are $\binom{L}{\ell}$ subsets $\mathcal{S} \subseteq \mathcal{L}$ such that $|\mathcal{S}| = \ell$, and by using the assumption that the SNRs at the antennas are i.i.d.

Using $\Pr\{\Upsilon\}$, we obtain $D(2,1,L)$ as given in (7).

b) $D(2,2,L)$: We derive the probability that the reference packet is successfully decoded in the second threshold-based decoding step. This probability is expressed as

$$\begin{aligned}D(2,2,L)=&\frac{1}{2}\big(\Pr\{\Delta\}+\Pr\{\Xi\}\big)\\=&\frac{1}{2}\big\{1+2(1\text{–}2P_{\mathsf{A}}\text{–}2P_{\mathsf{B}}\text{–}2P_{\mathsf{C}}\text{–}P_{\mathsf{D}})^L\\&\text{–}2(1\text{–}2P_{\mathsf{A}}\text{–}P_{\mathsf{B}}\text{–}2P_{\mathsf{C}}\text{–}P_{\mathsf{D}})^L\text{–}(1\text{–}2P_{\mathsf{A}}\text{–}2P_{\mathsf{B}})^L\big\}. \quad (9)\end{aligned}$$

Here, $\Pr\{\Delta\}$ denotes the probability that the two packets are decoded at the same antenna in $\mathcal{L}$. In contrast, $\Pr\{\Xi\}$ represents the probability that the packets are decoded across at least two distinct antennas but cannot both be decoded at any single antenna. By symmetry, the reference packet decoded in the second step is either P_k or P_l with probability $1/2$. In the following, we derive these two probabilities.

We first consider the case where both packets are decoded at a single antenna $i \in \mathcal{L}$. The probability of this event is given by $\Pr\{\Delta_i\} = \Pr\{\mathsf{A}_i\} + \Pr\{\breve{\mathsf{A}}_i\} = 2P_{\mathsf{A}}$. Consequently, the probability that P_k and P_l are successfully decoded at any antenna in $\mathcal{L}$ is $\Pr\{\Delta\} = \Pr\{\bigcup_{i\in\mathcal{L}} \Delta_i\}$. Using the inclusion-exclusion principle (5) and the binomial theorem, we obtain

$$\Pr\{\Delta\} = \Pr\{\textstyle\bigcup_{i\in\mathcal{L}} \Delta_i\} = 1-(1-2P_{\mathsf{A}})^L.$$

The derivation is analogous to (8) and is omitted for brevity.

Next, we derive the probability $\Pr\{\Xi\}$ that the two packets are decoded across at least two distinct antennas. To this end, consider the event $\Xi_{i,j,p}^{\mathsf{P}_l\to\mathsf{P}_k}$, which represents the situation where only P_l is independently decodable at Ant# i, P_k is decodable or in a PD state at Ant# j, and P_l and P_k cannot be simultaneously decoded at Ant# p.

This configuration accounts for the following two possible decoding mechanisms: a) P_l and P_k are separately decoded at Ant# i and Ant# j, respectively, and b) the initial decoding of P_l at Ant# i, followed by its subsequent removal at Ant# j through inter-antenna SIC, enables P_k to be decoded.

For distinct i, j, and p, this event is expressed as follows:

$$\Xi_{i,j,p}^{\mathsf{P}_l\to\mathsf{P}_k} = \breve{\mathsf{B}}_i \cap \{\mathsf{B}_j \cup \mathsf{C}_j \cup \breve{\mathsf{C}}_j \cup \mathsf{D}_j\} \cap \{\overline{\mathsf{A}_p \cup \breve{\mathsf{A}}_p}\}. \quad (10)$$

Define $\mathcal{L}_1 \subset \mathcal{L}$ as the subset of antenna elements where only P_l is independently decodable. Define $\mathcal{L}_2 \subseteq \mathcal{L}\backslash\mathcal{L}_1$ as the subset where P_k is decodable or in a PD state within each antenna. Let $\mathcal{L}_3 = \mathcal{L}\backslash(\mathcal{L}_1\cup\mathcal{L}_2)$ denote the subset comprising the remaining antenna elements at which P_l and P_k cannot be simultaneously decoded. Let $\ell_n = |\mathcal{L}_n|$, $n = 1,2,3$. Note that $\mathcal{L}_1\cup\mathcal{L}_2\cup\mathcal{L}_3 = \mathcal{L}$ and $\mathcal{L}_{n_1}\cap\mathcal{L}_{n_2} = \emptyset$ for $n_1 \neq n_2$.

For any $i \in \mathcal{L}_1$, $j \in \mathcal{L}_2$, and $p \in \mathcal{L}_3$, the intersection of sets (10) is written as

$$\begin{aligned}&\Pr\Big\{\bigcap_{\substack{i\in\mathcal{L}_1,j\in\mathcal{L}_2,\\ p\in\mathcal{L}\backslash(\mathcal{L}_1\cup\mathcal{L}_2)}} \Xi_{i,j,p}^{\mathsf{P}_l\to\mathsf{P}_k}\Big\}\\&=\prod_{i\in\mathcal{L}_1}\Pr\{\breve{\mathsf{B}}_i\}\prod_{j\in\mathcal{L}_2}\Pr\{\mathsf{B}_j\cup\mathsf{C}_j\cup\breve{\mathsf{C}}_j\cup\mathsf{D}_j\}\prod_{p\in\mathcal{L}\backslash(\mathcal{L}_1\cup\mathcal{L}_2)}\Pr\{\overline{\mathsf{A}_p\cup\breve{\mathsf{A}}_p}\}\\&=P_{\mathsf{B}}^{\ell_1}(P_{\mathsf{B}}+2P_{\mathsf{C}}+P_{\mathsf{D}})^{\ell_2}(1-2P_{\mathsf{A}})^{L-\ell_1-\ell_2}\\&\triangleq\Pr\{\Xi_{\mathcal{L}_1\mathcal{L}_2}^{\mathsf{P}_l\to\mathsf{P}_k}\}.\end{aligned}$$

Using the inclusion-exclusion principle (5), we have

$$\begin{aligned}\Pr\{\Xi^{\mathsf{P}_l\to\mathsf{P}_k}\}&=\Pr\Big\{\bigcup_{\substack{i,j,p\in\mathcal{L},\\ i\neq j\neq p}}\Xi_{i,j,p}^{\mathsf{P}_l\to\mathsf{P}_k}\Big\}\\&=\sum_{\substack{\mathcal{L}_1\subset\mathcal{L}\\ \mathcal{L}_1\neq\emptyset}}\sum_{\substack{\mathcal{L}_2\subseteq\mathcal{L}\backslash\mathcal{L}_1\\ \mathcal{L}_2\neq\emptyset}}(-1)^{|\mathcal{L}_1|+|\mathcal{L}_2|+2}\Pr\Big\{\bigcap_{\substack{i\in\mathcal{L}_1,j\in\mathcal{L}_2,\\ p\in\mathcal{L}\backslash(\mathcal{L}_1\cup\mathcal{L}_2)}}\Xi_{i,j,p}^{\mathsf{P}_l\to\mathsf{P}_k}\Big\}\\&=\sum_{\ell_1=1}^{L-1}\sum_{\ell_2=1}^{L-\ell_1}(-1)^{\ell_1+\ell_2}\sum_{\substack{\mathcal{L}_1\subset\mathcal{L}\\ |\mathcal{L}_1|=\ell_1}}\sum_{\substack{\mathcal{L}_2\subseteq\mathcal{L}\backslash\mathcal{L}_1\\ |\mathcal{L}_2|=\ell_2}}\Pr\Big\{\bigcap_{\substack{i\in\mathcal{L}_1,j\in\mathcal{L}_2,\\ p\in\mathcal{L}\backslash(\mathcal{L}_1\cup\mathcal{L}_2)}}\Xi_{i,j,p}^{\mathsf{P}_l\to\mathsf{P}_k}\Big\}\\&=\sum_{\ell_1=1}^{L-1}\sum_{\ell_2=1}^{L-\ell_1}(-1)^{\ell_1+\ell_2}\binom{L}{\ell_1}\binom{L-\ell_1}{\ell_2}\Pr\{\Xi_{\mathcal{L}_1\mathcal{L}_2}^{\mathsf{P}_l\to\mathsf{P}_k}\}.\end{aligned}$$

The term $\binom{L}{\ell_1}\binom{L-\ell_1}{\ell_2}$ accounts for the number of distinct antenna pairs $(\mathcal{L}_1,\mathcal{L}_2)$ of sizes ℓ_1 and ℓ_2, respectively, such that $\mathcal{L}_1$ and $\mathcal{L}_2$ are disjoint subsets of $\mathcal{L}$.

The symmetry of the SNRs for P_k and P_l ensures that $\Pr\{\Xi^{\mathsf{P}_l\leftarrow\mathsf{P}_k}\} = \Pr\{\Xi^{\mathsf{P}_l\to\mathsf{P}_k}\}$. The probability that P_l and P_k are decoded at the antennas in $\mathcal{L}_1$ and $\mathcal{L}_2$, respectively, is $\Pr\{\Xi_{\mathcal{L}_1\mathcal{L}_2}^{\mathsf{P}_l\to\mathsf{P}_k}\cap\Xi_{\mathcal{L}_1\mathcal{L}_2}^{\mathsf{P}_l\leftarrow\mathsf{P}_k}\} = P_{\mathsf{B}}^{\ell_1+\ell_2}(1-2P_{\mathsf{A}})^{L-\ell_1-\ell_2}$.

Therefore, we have

$$\begin{aligned}\Pr\{\Xi\}&=\Pr\{\Xi^{\mathsf{P}_l\to\mathsf{P}_k}\}+\Pr\{\Xi^{\mathsf{P}_l\leftarrow\mathsf{P}_k}\}-\Pr\{\Xi^{\mathsf{P}_l\to\mathsf{P}_k}\cap\Xi^{\mathsf{P}_l\leftarrow\mathsf{P}_k}\}\\&=\sum_{\ell_1=1}^{L-1}\sum_{\ell_2=1}^{L-\ell_1}(-1)^{\ell_1+\ell_2}\binom{L}{\ell_1}\binom{L-\ell_1}{\ell_2}\xi_{\Xi}(\ell_1,\ell_2),\end{aligned}$$

where

$$\xi_{\Xi}(\ell_1,\ell_2)=(2P_{\mathsf{B}}^{\ell_1}(P_{\mathsf{B}}+2P_{\mathsf{C}}+P_{\mathsf{D}})^{\ell_2}-P_{\mathsf{B}}^{\ell_1+\ell_2})(1-2P_{\mathsf{A}})^{L-\ell_1-\ell_2}.$$

Define $\ell_{12}\triangleq\ell_1+\ell_2$ with $\ell_{12}\in\{2,3,\ldots,L\}$. It follows that $\binom{L}{\ell_1}\binom{L-\ell_1}{\ell_2}=\binom{L}{\ell_{12}}\binom{\ell_{12}}{\ell_1}$. Thus, $\Pr\{\Xi\}$ can be rewritten as

$$\begin{aligned}\Pr\{\Xi\}&=\sum_{\ell_1=1}^{L-1}\sum_{\ell_{12}=\ell_1+1}^{L}(-1)^{\ell_{12}}\binom{L}{\ell_{12}}\binom{\ell_{12}}{\ell_1}\xi_{\Xi}(\ell_1,\ell_{12}-\ell_1)\\&=\sum_{\ell_{12}=2}^{L}\sum_{\ell_1=1}^{\ell_{12}-1}(-1)^{\ell_{12}}\binom{L}{\ell_{12}}\binom{\ell_{12}}{\ell_1}\\&\quad\cdot(2P_{\mathsf{B}}^{\ell_1}(P_{\mathsf{B}}+2P_{\mathsf{C}}+P_{\mathsf{D}})^{\ell_{12}-\ell_1}-P_{\mathsf{B}}^{\ell_{12}})(1-2P_{\mathsf{A}})^{L-\ell_{12}}.\end{aligned}$$

Finally, by applying the binomial theorem, we obtain

$$\begin{aligned}\Pr\{\Xi\}=&2(1\text{–}2P_{\mathsf{A}}\text{–}2P_{\mathsf{B}}\text{–}2P_{\mathsf{C}}\text{–}P_{\mathsf{D}})^L-2(1\text{–}2P_{\mathsf{A}}\text{–}P_{\mathsf{B}}\text{–}2P_{\mathsf{C}}\text{–}P_{\mathsf{D}})^L\\&\text{–}(1\text{–}2P_{\mathsf{A}}\text{–}2P_{\mathsf{B}})^L+(1\text{–}2P_{\mathsf{A}})^L.\end{aligned}$$

With $\Pr\{\Delta\}$ and $\Pr\{\Xi\}$, we obtain $D(2,2,L)$ in (9).

Remark 1. *For the single-antenna case $L = 1$, $R \geq 1$, and $m \leq 2$, the average decoding error probability $\bar{\epsilon}_{m,1}$ in* (4) *reduces to the result in [5]. A contribution of this work is the derivation of the exact average decoding error probability for the broader range $R > 0$ with $m \leq 2$, by characterizing the integration regions for $0 < R < 1$ as detailed in Sec. II-B2.*

B. Closed-Form Approximation for $m \geq 3$

For $m \geq 3$, the decoding condition at each SIC step depends on preceding SIC outcomes because a packet decoded at one antenna is canceled from all antenna observations through

inter-antenna SIC. While exact expressions for $m \leq 2$ can be derived by enumerating the decoding events and applying the inclusion-exclusion principle, the number of joint decoding-event patterns grows rapidly with m and L, making a direct extension analytically intractable.

To derive a closed-form approximation for $D(m, L)$, let $\mathcal{A}_i$ denote the event that the ith non-reference packet is decoded at the first SIC step, and let $\mathcal{B}_i$ denote the event that the reference packet is eventually decoded after $\mathcal{A}_i$ occurs. Conditioning on the packet decoded at the first SIC step yields

$$D(m, L) = D(m, 1, L) + (m-1) D(m, 1, L) \Pr(\mathcal{B}_i \mid \mathcal{A}_i). \quad (11)$$

The first term represents the probability that the reference packet is decoded at the first SIC step. The second term represents the probability that one of the $m-1$ non-reference packets is decoded first and canceled, after which the reference packet is decoded from the residual $(m-1)$-packet collision.

To obtain a tractable approximation, after each cancellation, the residual collision is modeled as an independent collision with the same statistical characteristics as a newly generated collision consisting of the remaining packets. Hence,

$$\Pr(\mathcal{B}_i \mid \mathcal{A}_i) \approx \widetilde{D}(m-1, L), \quad (12)$$

where $\widetilde{D}(m, L)$ denotes the approximate probability that the reference packet is eventually decoded from an m-packet collision. Substituting (12) into (11) gives

$$\begin{aligned} D(m, L) &\approx \widetilde{D}(m, L) \\ &= D(m, 1, L) \left(1 + (m-1)\widetilde{D}(m-1, L)\right), \ m \geq 3. \end{aligned}$$

Accordingly, $\widetilde{D}(m, L)$ is recursively defined as

$$\widetilde{D}(m, L) = \begin{cases} D(2, L), & m = 2, \\ D(m, 1, L)\big(1 + (m-1)\widetilde{D}(m-1, L)\big), & m \geq 3, \end{cases} \quad (13)$$

where the exact two-packet decoding probability $D(2, L) = D(2, 1, L) + D(2, 2, L)$ is obtained from (7) and (9). By recursively expanding (13), $\widetilde{D}(m, L)$ can equivalently be expressed in the following closed form:

$$\begin{aligned} \widetilde{D}(m, L) = &\sum_{j=3}^{m} \tfrac{(m-1)!}{(j-1)!} \prod_{r=j}^{m} D(r, 1, L) \\ &+ (m-1)! D(2, L) \prod_{r=3}^{m} D(r, 1, L), \ \ m \geq 3. \end{aligned} \quad (14)$$

The term $D(r, 1, L)$ denotes the first-step decoding probability for an arbitrary collision size r. Since its exact expression is derived in Appendix II, the approximation above can be evaluated over the entire range $R > 0$.

Finally, for $m \geq 3$, $\bar{\epsilon}_{m,L}$ is approximated as

$$\bar{\epsilon}_{m,L} \approx 1 - \widetilde{D}(m, L). \quad (15)$$

Remark 2 (Conventional SA)**.** *The average decoding error probability in* (4) *for* $m \leq 2$ *and its approximation in* (15) *for* $m \geq 3$ *characterize the multi-antenna SC-SIC receiver within a slot. These results are directly applicable to conventional SA with L-antenna reception, in which packet repetition and inter-slot SIC are not employed. In particular, for normalized offered traffic G, the PLR of conventional SA is*

$$\mathrm{PLR}^{\mathrm{SA}}(G, L, \bar{\gamma}, \eta_0) = \sum_{m=1}^{\infty} e^{-G} \tfrac{G^{m-1}}{(m-1)!} \bar{\epsilon}_{m,L}, \quad (16)$$

TABLE III

Average decoding error probability $\bar{\epsilon}_{m,L}$ $(m = 1, 2, 3, 4)$ within a slot. "Gap" denotes the difference $|\text{Sim.} - \text{Theo.}|$. Simulation results reported as $< 10^{-6}$ indicate that no decoding error was observed over 10^6 trials. $(\eta_0 = 10, \bar{\gamma} = 10$ dB).

m	L	MRC-SIC [4, Theorem 3]			SC-SIC based on (4) or (15)		
		Sim.	Theo.	Gap	Sim.	Theo.	Gap
1	4	1.88×10^{-2}	1.90×10^{-2}	1.7×10^{-4}	1.60×10^{-1}	1.60×10^{-1}	1.0×10^{-4}
	8	1.00×10^{-5}	1.00×10^{-5}	$< 10^{-6}$	2.53×10^{-2}	2.55×10^{-2}	2.0×10^{-4}
	16	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	7.00×10^{-4}	7.20×10^{-4}	1.8×10^{-5}
	32	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$
	64	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$
2	4	6.16×10^{-1}	7.32×10^{-1}	1.2×10^{-1}	7.89×10^{-1}	7.89×10^{-1}	5.1×10^{-5}
	8	3.68×10^{-1}	4.81×10^{-1}	1.1×10^{-1}	5.84×10^{-1}	5.84×10^{-1}	9.7×10^{-5}
	16	1.41×10^{-1}	2.07×10^{-1}	6.6×10^{-2}	3.31×10^{-1}	3.31×10^{-1}	1.7×10^{-4}
	32	2.25×10^{-2}	3.84×10^{-2}	1.6×10^{-2}	1.10×10^{-1}	1.09×10^{-1}	3.9×10^{-4}
	64	7.00×10^{-4}	1.30×10^{-3}	6.0×10^{-4}	1.19×10^{-2}	1.19×10^{-2}	3.1×10^{-5}
3	4	9.38×10^{-1}	9.20×10^{-1}	1.8×10^{-2}	9.85×10^{-1}	9.83×10^{-1}	1.7×10^{-3}
	8	7.53×10^{-1}	8.31×10^{-1}	7.8×10^{-2}	9.60×10^{-1}	9.57×10^{-1}	2.8×10^{-3}
	16	3.82×10^{-1}	6.20×10^{-1}	2.4×10^{-1}	8.97×10^{-1}	8.94×10^{-1}	3.5×10^{-3}
	32	6.85×10^{-2}	1.92×10^{-1}	1.2×10^{-1}	7.67×10^{-1}	7.64×10^{-1}	2.7×10^{-3}
	64	7.93×10^{-4}	1.66×10^{-3}	8.6×10^{-4}	5.61×10^{-1}	5.60×10^{-1}	1.1×10^{-3}
4	4	9.92×10^{-1}	9.68×10^{-1}	2.4×10^{-2}	9.99×10^{-1}	9.99×10^{-1}	2.1×10^{-5}
	8	9.43×10^{-1}	9.23×10^{-1}	2.0×10^{-2}	9.98×10^{-1}	9.98×10^{-1}	6.9×10^{-5}
	16	6.76×10^{-1}	8.05×10^{-1}	1.3×10^{-1}	9.94×10^{-1}	9.94×10^{-1}	2.6×10^{-4}
	32	1.61×10^{-1}	4.53×10^{-1}	2.9×10^{-1}	9.86×10^{-1}	9.85×10^{-1}	4.8×10^{-4}
	64	3.39×10^{-2}	2.32×10^{-2}	1.1×10^{-2}	9.62×10^{-1}	9.60×10^{-1}	1.1×10^{-3}

and the throughput is given by $T^{\mathrm{SA}}(G, L, \bar{\gamma}, \eta_0) = G[1 - \mathrm{PLR}^{\mathrm{SA}}(G, L, \bar{\gamma}, \eta_0)]$*, where* $m-1$ *is the number of packets interfering with the reference packet. A detailed performance evaluation of conventional SA is omitted due to space limitations, as the present work focuses on IR-SA.*

Remark 3. *For* $m = 2$*,* $\bar{\epsilon}_{2,L}$ *of MRC-SIC in [4, Theorem 3] is obtained using an approximation whose accuracy improves in the large-L regime. In contrast, the SC-SIC analysis yields an exact closed-form expression for* $\bar{\epsilon}_{2,L}$ *for any finite L. For* $m \geq 3$*, the proposed approximation explicitly accounts for decoding of the reference packet after preceding interference cancellations, while the normal approximation in MRC-SIC [4, Theorem 3] does not explicitly represent subsequent SIC steps. As shown in Table III, the exact result for* $m = 2$ *and the approximation for* $m \geq 3$ *agree closely with the SC-SIC simulations over the considered finite values of L, whereas the MRC-SIC approximations show a noticeable mismatch.*

C. High-SNR Limits for Finite m *and* L *(*$\bar{\gamma} \to \infty$*)*

In this subsection, we characterize the high-SNR behavior of $\bar{\epsilon}_{m,L}$ for a fixed finite m and L. We first establish the conditions under which $\bar{\epsilon}_{m,L}$ vanishes as $\bar{\gamma} \to \infty$. We then characterize the nonzero error floor that arises from residual multiuser interference when this condition is not satisfied.

We first consider a singleton slot. Here, the average decoding error probability vanishes in the high-SNR regime, i.e.,

$$\lim_{\bar{\gamma} \to \infty} \bar{\epsilon}_{1,L} = 0. \quad (17)$$

This result follows directly from (6).

We now turn to an m-packet collision with $m \geq 2$. Let $\gamma_{1,i} > \gamma_{2,i} > \cdots > \gamma_{m,i}$ denote the ordered received SNRs at Ant# i. The SINR of the strongest packet satisfies

$$\eta_{1,i} = \frac{\gamma_{1,i}}{1 + \sum_{k=2}^{m} \gamma_{k,i}} > \frac{\gamma_{1,i}}{1 + (m-1)\gamma_{1,i}}. \quad (18)$$

As $\bar{\gamma} \to \infty$, the lower bound converges to

$$\lim_{\bar{\gamma}\to\infty} \frac{\gamma_{1,i}}{1+(m-1)\gamma_{1,i}} = \frac{1}{m-1}. \tag{19}$$

For $\eta_0 < 1/(m-1)$, (18) and (19) ensure that the strongest packet is decodable at high SNR. At equality, the fading gains are distinct almost surely, so the aggregate interference is strictly less than $(m-1)$ times the strongest gain and its limiting SINR exceeds $1/(m-1)$. Thus, $\eta_0 \leq 1/(m-1)$ permits SIC to be initiated almost surely. Applying the same argument after each cancellation decodes all m packets, and hence no high-SNR error floor remains.

In contrast, suppose that $\eta_0 > 1/(m-1)$. At the equal-power point, the first-step SINR of every packet at every antenna equals $1/(m-1)$. By continuity, there exists a neighborhood of this point in which none of the packets is decodable at any antenna. Because this neighborhood has positive probability under continuous fading, SIC fails with positive probability, resulting in a nonzero high-SNR error floor.

Combining the above two cases, for $m \geq 2$ and fixed finite L, the high-SNR behavior of $\bar{\epsilon}_{m,L}$ is characterized by

$$\lim_{\bar{\gamma}\to\infty} \bar{\epsilon}_{m,L} = \begin{cases} 0, & 0 < \eta_0 \leq \frac{1}{m-1}, \\ \bar{\epsilon}_{m,L}^{\text{floor}}(\eta_0) > 0, & \eta_0 > \frac{1}{m-1}. \end{cases} \tag{20}$$

Since $R = \log_2(1+\eta_0)$, the *collision-order-dependent spectral-efficiency condition* for a vanishing high-SNR error floor is $R \leq \log_2\left(\frac{m}{m-1}\right)$. For higher-order collisions ($m \geq 3$), this condition requires $R < 1$, and its upper bound decreases as m increases.

We next characterize the value of the error floor in the regime $\eta_0 > 1/(m-1)$. For $m = 2$, the error floor is obtained exactly. For $m \geq 3$, it is approximated by taking the high-SNR limit of the proposed approximation:

$$\bar{\epsilon}_{2,L}^{\text{floor}}(\eta_0) = \left(\tfrac{\eta_0-1}{\eta_0+1}\right)^L, \tag{21}$$

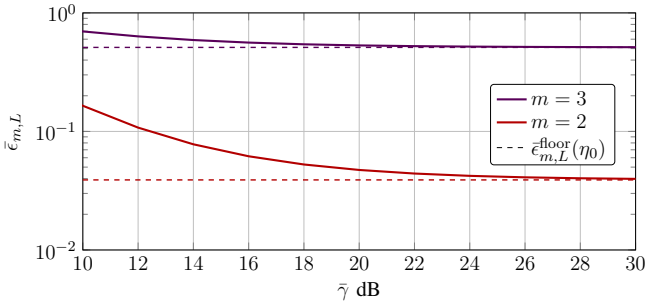
$$\bar{\epsilon}_{m,L}^{\text{floor}}(\eta_0) \approx 1 - \frac{1}{m}\left[\sum_{j=3}^{m}\prod_{r=j}^{m}\Phi_{r,L}(\eta_0) + 2\prod_{r=2}^{m}\Phi_{r,L}(\eta_0)\right], \tag{22}$$

where $m \geq 3$, $\eta_0 > 1/(m-1)$, and

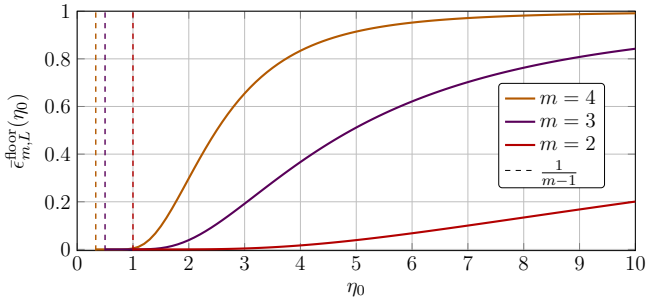
$$\Phi_{r,L}(\eta_0) = 1 - \left[1 - \sum_{\substack{1\leq s\leq r \\ 1-(s-1)\eta_0>0}} (-1)^{s+1}\binom{r}{s}\left(\frac{1-(s-1)\eta_0}{1+\eta_0}\right)^{r-1}\right]^L. \tag{23}$$

Detailed derivations of the exact result in (21) and the approximate result in (22) are provided in Appendix III. The impact of these slot-level error floors on the system-level BP thresholds is illustrated later by the high-SNR benchmarks in Fig. 8(b).

Figure 5(a) shows that $\bar{\epsilon}_{2,L}$ approaches the exact high-SNR floors in (21) as $\bar{\gamma}$ increases. Figure 5(b) plots the exact and approximate floors in (21) and (22), respectively, versus η_0 for $m = 2, 3, 4$. The vertical dashed lines mark $\eta_0 = 1/(m-1)$, above which the floor is positive and increases more rapidly for larger m.



(a) $\bar{\epsilon}_{m,L}$ versus the average received SNR $\bar{\gamma}$ for $m = 2, 3$ ($L = 8$, $\eta_0 = 5$).



(b) High-SNR error floor $\bar{\epsilon}_{m,L}^{\text{floor}}$ versus the decoding threshold η_0 for $m = 2, 3$, and 4 ($L = 8$). Only the nonzero regimes $\eta_0 > 1/(m-1)$ are shown; the vertical dashed lines mark the corresponding boundaries.

Fig. 5: High-SNR behavior of the average decoding error probability and its error floor (exact for $m = 2$ and approximate for $m = 3, 4$).

IV. IR-SA System Model

A. IR-SA Access Protocol with Multi-Antenna Reception

We consider an IR-SA system with multi-antenna reception, where N_{T} single-antenna devices communicate with an AP equipped with L receive antennas over a frame of M time slots. The antennas at the AP are spaced sufficiently far apart to ensure independent fading values. In each frame, each device becomes active independently with probability $\pi \ll 1$ and, if active, generates one message that is encoded and modulated into a packet at a spectral efficiency R using a common channel code $\mathcal{C}$ and a modulation scheme. Each packet, and hence each of its replicas, occupies one slot. Each active device selects a repetition degree l with probability Λ_l according to the distribution $\{\Lambda_l\}_{l=1}^{l_{\max}}$. It transmits l replicas of its packet in l randomly selected slots within a frame.

We define the normalized population size as $\alpha \triangleq N_{\text{T}}/M$, which is held constant in the asymptotic regime. Since $\mathbb{E}[N_{\text{a}}] = \pi N_{\text{T}}$, the normalized offered traffic is defined as $G \triangleq \frac{\mathbb{E}[N_{\text{a}}]}{M} = \pi\alpha$, which represents the average number of packets transmitted per slot.

As illustrated in Fig. 2, the multi-antenna IR-SA protocol outlined in Sec. IV-A can be characterized in a manner analogous to single-antenna IR-SA [6], [15]. Specifically, it is modeled as a bipartite graph $\mathcal{G} = (\mathcal{D}, \mathcal{S}, \mathcal{E})$. Here, $\mathcal{D}$ denotes the set of N_{T} device nodes (DNs), $\mathcal{S}$ denotes the set of M slot nodes (SNs), and $\mathcal{E}$ represents the set of edges. An edge connects a DN $d_i \in \mathcal{D}$ to an SN $s_j \in \mathcal{S}$ if and only if the i-th device selects the j-th slot for transmission.

In the considered receiver, SC-SIC processing across the L antennas is confined to each slot and therefore does not

alter the underlying bipartite graph. Following the slot-level abstraction in [4], when m packets remain unresolved in a slot, the effect of multi-antenna processing is represented by the reference-packet decoding success probability $1-\bar{\epsilon}_{m,L}$. Consequently, multi-antenna reception changes only the SN decoding rule, while the graph-level DE recursion retains the same structural form as in single-antenna IR-SA.

The number of edges connected to a DN or SN is defined as the node degree. In this framework, a DN has degree l (i.e., it has l edges connected to SNs) with probability Λ_l. The set $\{\Lambda_l\}_{l=1}^{l_{\max}}$ is the DN degree distribution, and the average number of replicas per active device is given by $\bar{d} \triangleq \Lambda'(1) = \sum_{l=1}^{l_{\max}} l\Lambda_l$. As $N_{\mathrm{T}} \to \infty$, the SN degree distribution is Poisson with $\Psi_n = \exp(-\alpha\bar{d})(\alpha\bar{d})^n/n!$ [6], [15].

The degree distributions are also defined from an edge-perspective in polynomials [15]: $\lambda(x) = \sum_{l=1}^{l_{\max}} \lambda_l x^{l-1}$ with $\lambda_l = \frac{l\Lambda_l}{\sum_{l=1}^{l_{\max}} l\Lambda_l}$ and

$$\rho(x) = \textstyle\sum_{n=1}^{N_{\mathrm{T}}} \rho_n x^{n-1} = \exp(-\alpha\bar{d}(1-x)), \tag{24}$$

with Poisson distribution

$$\rho_n = \frac{n\Psi_n}{\sum_{n=1}^{N_{\mathrm{T}}} n\Psi_n} = \exp(-\alpha\bar{d})\frac{(\alpha\bar{d})^{n-1}}{(n-1)!},\ n = 1, 2, \ldots. \tag{25}$$

B. Spatio-Temporal SIC Decoding in Rayleigh Block Fading

The AP follows the standard iterative IR-SA decoding procedure in [6], [7]. In each iteration, the intra-slot SC-SIC procedure described in Sec. II-A is applied to each unresolved slot. Once a packet is decoded, its remaining replicas are identified and canceled from all L antenna observations in the corresponding slots through *inter-slot SIC*. The affected slots are then processed again, and this procedure is repeated until no additional packet is decoded. In the bipartite graph $\mathcal{G}$, these operations correspond to removing all edges incident to a successfully decoded DN, thereby reducing the residual degrees of the connected SNs.

V. Density Evolution with L-Antenna Reception

In this section, building on the DE framework for IR-SA [5]–[7], [17], we incorporate the L-antenna decoding probabilities from Sec. III into the SN update and characterize the asymptotic performance of multi-antenna IR-SA.

We refer to the DNs (SNs) collectively as the DN (SN) group. In the protograph, let ${}^{\ell}p$ denote the probability that an edge incident on the DN group carries an erasure message toward the SN group at iteration ℓ. Similarly, let ${}^{\ell}q$ denote the probability that an edge incident on the SN group carries an erasure message toward the DN group.

First, consider the DN group with degree distribution $\lambda(x)$. At iteration ℓ, an edge incident to a degree-l DN is removed if at least one of the other $l-1$ edges has been successfully resolved. Therefore, given the device activation probability π, the average probability that a DN-to-SN message remains erased is given by [6], [7], [17]

$${}^{\ell}p = \pi \textstyle\sum_{l=1}^{l_{\max}} \lambda_l \cdot {}^{(\ell-1)}q^{l-1} = \pi\lambda({}^{(\ell-1)}q). \tag{26}$$

Second, consider the SN group with degree distribution $\rho(x)$. The average probability that an SN-to-DN message remains erased at iteration ℓ is given by ${}^{\ell}q = \sum_{r=1}^{\infty} \rho_r \cdot {}^{\ell}q^{(r)}$, where ρ_r is the fraction of edges connected to a degree-r SN (see (25)), and ${}^{\ell}q^{(r)}$ denotes the conditional erasure probability of an outgoing message from a degree-r SN. Following the single-antenna DE formulation [5], [7] and the multi-antenna slot-level abstraction [4], we obtain

$${}^{\ell}q^{(r)} = 1 - \textstyle\sum_{m=1}^{r} \binom{r-1}{m-1}(1-{}^{\ell}p)^{r-m}({}^{\ell}p)^{m-1}(1-\bar{\epsilon}_{m,L}). \tag{27}$$

For each $m \in \{1, \ldots, r\}$, the factor $\binom{r-1}{m-1}(1-{}^{\ell}p)^{r-m}({}^{\ell}p)^{m-1}$ is the probability that, among the other $r-1$ replicas incident on a degree-r SN, $r-m$ replicas have already been removed through inter-slot SIC, while $m-1$ replicas remain unresolved. Together with the reference replica, these unresolved replicas give an SN residual degree of m. Conditioned on this residual degree, $1-\bar{\epsilon}_{m,L}$ is the probability that the reference replica is successfully decoded. For DE, all effects of the L-antenna receiver, including packet–antenna selection and intra- and inter-antenna SIC, are *encapsulated* in slot-level decoding probability $1-\bar{\epsilon}_{m,L}$. Therefore, no antenna-specific messages need to be introduced into the bipartite graph. The corresponding $\bar{\epsilon}_{m,L}$ is derived in Sec. III.

Once the multi-antenna receiver processing is encapsulated in $1-\bar{\epsilon}_{m,L}$, the subsequent graph-level derivation follows the single-antenna DE framework in [5], [7]. Averaging over the SN degree distribution, we obtain

$${}^{\ell}q = 1 - e^{-\alpha\bar{d}{}^{\ell}p} \textstyle\sum_{m=0}^{\infty} \frac{(\alpha\bar{d}{}^{\ell}p)^m}{m!}(1-\bar{\epsilon}_{m+1,L}). \tag{28}$$

From (24), we have $\alpha\bar{d}x = -\ln\rho(1-x)$, which allows ${}^{\ell}q$ to be expressed as a function of $(1-{}^{\ell}p)$:

$${}^{\ell}q = 1 - \chi_L(1-{}^{\ell}p), \tag{29}$$

where the L-antenna SN decoding function $\chi_L(\cdot)$ is given by

$$\chi_L(1-x) \triangleq e^{-\alpha\bar{d}x} \textstyle\sum_{m=0}^{\infty} \frac{(\alpha\bar{d}x)^m}{m!}(1-\bar{\epsilon}_{m+1,L}). \tag{30}$$

Accordingly, $\chi_L(\cdot)$ is the only component through which L-antenna reception enters the DE recursion, while the graph-level DE recursion retains the same form as in the single-antenna case.

Substituting (26) into (28) and using $G = \pi\alpha$, the DE recursion becomes, with ${}^{(\ell=0)}q = 1$,

$${}^{\ell}q = 1 - e^{-G\bar{d}\lambda({}^{(\ell-1)}q)} \textstyle\sum_{m=0}^{\infty} \frac{(G\bar{d}\lambda({}^{(\ell-1)}q))^m}{m!}(1-\bar{\epsilon}_{m+1,L}). \tag{31}$$

Let ${}^{(\infty)}q(G, \bar{d}, L, \bar{\gamma}, \eta_0) = \lim_{\ell} {}^{(\ell)}q$ denote the limiting erasure probability of the DE recursion. The corresponding asymptotic PLR is [5], [6]

$$\begin{aligned}\mathrm{PLR}(G, \bar{d}, L, \bar{\gamma}, \eta_0) &= \Lambda({}^{(\infty)}q(G, \bar{d}, L, \bar{\gamma}, \eta_0)) \\ &= \textstyle\sum_{l=1}^{l_{\max}} \Lambda_l \left({}^{(\infty)}q(G, \bar{d}, L, \bar{\gamma}, \eta_0)\right)^l. \end{aligned} \tag{32}$$

For a target PLR^*, the asymptotic iterative decoding threshold, also referred to as the BP decoding threshold, is defined as the supremum offered traffic [5], [7] for which the asymptotic PLR does not exceed PLR^*

$$G^{\mathrm{BP,blk}} = \sup\{G : \mathrm{PLR}(G, \bar{d}, L, \bar{\gamma}, \eta_0) \le \mathrm{PLR}^*\}. \tag{33}$$

Under the κ-MaxDecoding constraint defined in Remark 4, the sum in (31) is restricted to $m = 0, \ldots, \kappa - 1$, since the residual degree is $m + 1 \leq \kappa$.

Remark 4 (κ-MaxDecoding)**.** *In* (31)*, the summation extends to infinity due to the unbounded support of the Poisson distribution. However, practical systems typically restrict SIC to low-order collisions because of interference, error propagation, and computational complexity [31]. Accordingly, we adopt κ-MaxDecoding, in which intra-slot SIC in Sec. II-A is performed only when the number of residual packets in a slot does not exceed κ. A slot whose initial collision degree exceeds κ can nevertheless become eligible for intra-slot SIC after inter-slot SIC reduces its residual degree to at most κ.*

For fixed finite L and $\kappa \geq 2$, $m = \kappa$ is the most restrictive among the processed residual collision degrees $2 \leq m \leq \kappa$. Hence, the high-SNR error floor vanishes for every processed residual collision degree if and only if $R \leq \log_2(\kappa/(\kappa-1))$.

VI. Converse Bound and MAP Threshold

We investigate fundamental limits on asymptotically supportable offered traffic under the SC-SIC slot-node model through converse-bound and MAP-threshold analyses, extending the EXIT-area and trial-entropy framework in [7] using $\chi_L(\cdot)$ in (30).

A. Converse Bound

Following the EXIT-area argument for single-antenna IR-SA in [7, Sec. III-A], a necessary condition for successful iterative decoding is that the DN and SN EXIT functions form an open decoding tunnel. Suppressing the iteration index in (26) and (29), the corresponding transfer functions for the present multi-antenna system are

$$p = \pi\lambda(q) \triangleq f_{\mathrm{d}}(q), \qquad q \in (0, 1], \tag{34}$$

$$q = 1 - \chi_L(1-p) \triangleq f_{\mathrm{s}}(p), \qquad p \in (0, \pi]. \tag{35}$$

Thus, the effect of L-antenna reception is incorporated into the EXIT analysis solely through $\chi_L(\cdot)$, defined in (30).

As established in [7], the open-tunnel condition requires [16], [21], [25]

$$A_{\mathrm{d}} + A_{\mathrm{s}} < \pi, \tag{36}$$

where, by applying the same EXIT-area argument as in [7] to the present DN and SN EXIT functions,

$$A_{\mathrm{d}} = \int_0^1 f_{\mathrm{d}}(q)\mathrm{d}q = \frac{\pi}{\bar{d}}, \text{ and } A_{\mathrm{s}} = \int_0^{\pi} f_{\mathrm{s}}(p)\mathrm{d}p = \pi - \tau_L(\pi), \tag{37}$$

with

$$\tau_L(x) \triangleq \int_0^x \chi_L(1-z)\mathrm{d}z. \tag{38}$$

Under the κ-MaxDecoding constraint in **Remark 4**, substituting (30) into (38) yields

$$\tau_L(x) = \frac{1}{\alpha\bar{d}} \sum_{m=0}^{\kappa-1} (1 - \bar{\epsilon}_{m+1,L}) \left[1 - e^{-\alpha\bar{d}x} \sum_{k=0}^{m} \frac{(\alpha\bar{d}x)^k}{k!}\right]. \tag{39}$$

Combining (36) and (37), the open-tunnel condition becomes $\pi\alpha < \alpha\bar{d}\,\tau_L(\pi)$. With $G = \pi\alpha$, the boundary of this condition, $\pi\alpha = \alpha\bar{d}\,\tau_L(\pi)$, is given by

$$G^{\mathrm{C,blk}} = \sum_{m=0}^{\kappa-1} (1 - \bar{\epsilon}_{m+1,L}) \left[1 - e^{-G^{\mathrm{C,blk}}\bar{d}} \sum_{k=0}^{m} \frac{(G^{\mathrm{C,blk}}\bar{d})^k}{k!}\right]. \tag{40}$$

The positive solution of (40) gives the converse bound $G^{\mathrm{C,blk}}$. For a fixed average repetition degree $\bar{d}$, $G^{\mathrm{C,blk}}$ is independent of the particular DN degree distribution $\Lambda(x)$.

Equation (40) extends the single-antenna converse bound in [7] to finite-L multi-antenna reception through the decoding success probability $1 - \bar{\epsilon}_{m+1,L}$.

B. MAP Threshold

Trial entropy, used to characterize MAP decoding of low-density parity-check (LDPC) codes over the binary erasure channel (BEC) [16], was subsequently applied to a regular-repetition SA scheme over the collision channel [17] and to single-antenna IR-SA over Rayleigh block-fading channels [7, Sec. III-B], where the device activation probability π plays a role analogous to the BEC erasure probability. Since finite-L multi-antenna reception is encapsulated in the SN decoding function $\chi_L(\cdot)$, we apply the same framework using $\chi_L(\cdot)$.

A nonzero DE fixed point x, representing the limiting DN-to-SN erasure probability x, satisfies the fixed-point equation

$$x = \pi\lambda\left(1 - \chi_L(1-x)\right). \tag{41}$$

Equivalently, the activation probability associated with x is

$$\pi_L(x) = \frac{x}{\lambda\left(1 - \chi_L(1-x)\right)}, \qquad x \in (0, \bar{x}], \tag{42}$$

where $\pi_L(\bar{x}) = 1$ and $0 < \bar{x} < 1$. The trial entropy associated with the parametric BP EXIT curve $\big(\pi_L(x), \Lambda(1-\chi_L(1-x))\big)$ is [16, Definition 3.119], [7, Eq. (36)]

$$\begin{aligned} P_L(x) &\triangleq \int_0^x \Lambda\left(1 - \chi_L(1-z)\right) \mathrm{d}\pi_L(z) \\ &= \frac{x\Lambda\left(1 - \chi_L(1-x)\right)}{\lambda\left(1 - \chi_L(1-x)\right)} + \Lambda'(1)\left[x\chi_L(1-x) - \tau_L(x)\right], \end{aligned}$$

where $\tau_L(x)$ is defined in (38).

Under the trial-entropy criterion, let x^{MAP} denote the nonzero root of $P_L(x^{\mathrm{MAP}}) = 0$ corresponding to the MAP transition. Using $G = \alpha\pi$, the MAP threshold is

$$G^{\mathrm{MAP,blk}} = \alpha\pi_L(x^{\mathrm{MAP}}). \tag{43}$$

For $L = 1$, this reduces to the single-antenna result in [7].

VII. Numerical and Simulation Results

In this section, by applying the average decoding error probabilities $\bar{\epsilon}_{m,L}$ derived in Sec. III, we evaluate the asymptotic performance of multi-antenna IR-SA.

Figure 6 presents the PLR of the SC-SIC-based IR-SA, evaluated using (32), for $L = 8$ and $\kappa = 2$. For comparison, the corresponding performance of multi-antenna IR-SA with MRC-SIC [4] and minimum mean-square error-based SIC (MMSE-SIC) [32] is also shown. For reference, the single-antenna IR-SA ($L = 1$) [5] is also included. The same degree

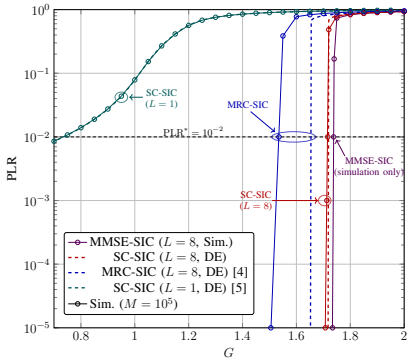


Fig. 6: PLRs vs. G of IR-SA for the SC-SIC, the MRC-SIC [4], and the MMSE-SIC (simulation only): $\kappa = 2$, $\eta_0 = 5$, $\bar{\gamma} = 20$ dB, and $\Lambda(x) = 0.586x^2 + 0.291x^3 + 0.031x^6 + 0.022x^7 + 0.006x^8 + 0.063x^{10}$.

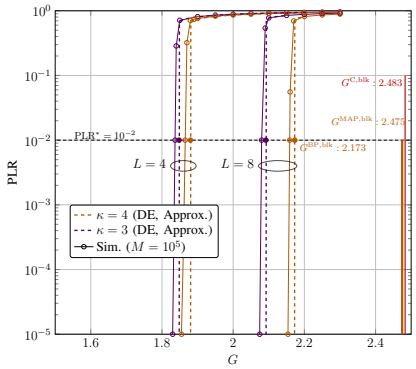


Fig. 7: PLRs vs. G of SC-SIC-based IR-SA obtained from DE using the proposed approximation and from simulations, for $\kappa = 3, 4$ and $L = 4, 8$. The BP threshold $G^{\mathrm{BP,blk}}$, MAP threshold $G^{\mathrm{MAP,blk}}$, and converse bound $G^{\mathrm{C,blk}}$ for $L = 8$ and $\kappa = 4$ are also indicated. Other parameters are as in Fig. 6.

distribution, optimized specifically for the considered multi-antenna IR-SA via a differential evolution algorithm [33], is applied to all schemes. Since no analytical result is available for MMSE-SIC, its performance is evaluated by simulation only, where threshold-based decoding is applied after MMSE filtering. Simulation results with a finite number of slots ($M = 10^5$) for SC-SIC closely match the asymptotic DE analysis, thereby validating our analytical framework. In contrast, the MRC-SIC approach in [4] exhibits a noticeable discrepancy between simulation and analysis. This discrepancy arises because the MRC-SIC analysis is approximate even for $m = 2$, whereas the SC-SIC analysis is exact for $m \leq 2$, as confirmed by the results in Table III.

In Fig. 6, the horizontal line $\mathrm{PLR}^* = 10^{-2}$ indicates the target used to define $G^{\mathrm{BP,blk}}$. Its intersections with the DE curves give the corresponding BP thresholds, whereas its intersection with the simulation-only MMSE-SIC curve provides only a finite-length simulation-based operating point.

Beyond the performance analysis, it is also important to consider the computational complexity of the multi-antenna processing in each decoding attempt. MMSE-SIC requires the inversion of an $L \times L$ interference-plus-noise covariance matrix for MMSE filtering, which generally entails cubic complexity with respect to L. In contrast, SC-SIC avoids such matrix operations and relies on per-antenna SINR evaluation and branch selection, whose complexity scales linearly with L. Both schemes additionally require packet reconstruction and interference cancellation as part of the iterative SIC procedure. For the parameter setting considered in Fig. 6, SC-SIC exhibits PLR performance close to that of MMSE-SIC while avoiding the computationally intensive matrix inversion.

Figure 7 compares the PLRs obtained from DE with the corresponding simulation results for SC-SIC-based IR-SA as functions of the offered traffic G, for $\kappa \in \{3, 4\}$ and $L \in \{4, 8\}$. The DE results agree well with the corresponding simulation results for all considered combinations of κ and L, demonstrating the accuracy of the proposed approximation. Compared to the exact agreement between the theoretical analysis and the simulation results for $\kappa = 2$ in Fig. 6, the slight remaining discrepancies here are attributable to the independence assumption in (12).

To position the BP performance relative to MAP decoding and the converse bound, Fig. 7 also shows $G^{\mathrm{BP,blk}} = 2.173$, $G^{\mathrm{MAP,blk}} = 2.475$, and $G^{\mathrm{C,blk}} = 2.483$ for $L = 8$ and $\kappa = 4$. The MAP threshold exceeds the BP threshold by 0.302 and lies only 0.008 below the converse bound, indicating that the bound is tight for this setting. The remaining BP-to-MAP gap motivates the spatial-coupling analysis presented below.

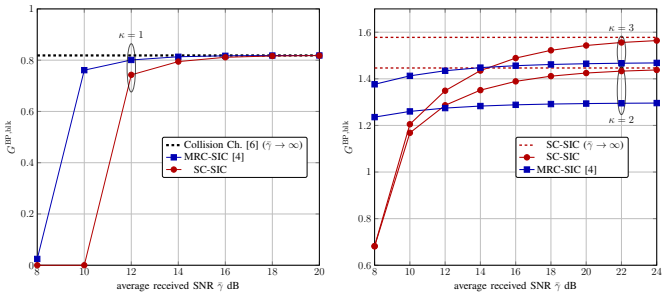
Figures 8(a) and 8(b) plot the BP thresholds $G^{\mathrm{BP,blk}}$ of MRC-SIC [4] and SC-SIC versus the average received SNR $\bar{\gamma}$ for $\kappa = 1$ and $\kappa = 2, 3$, respectively. Following [4], we set $\eta_0 = 10$, corresponding to $R = \log_2(11) \approx 3.46$ bit/s/Hz.

For $\kappa = 1$, MRC-SIC consistently outperforms SC-SIC at finite SNR because it maximizes the received SNR of the desired packet. As $\bar{\gamma} \to \infty$, however, the singleton decoding error probabilities of both schemes vanish, and their BP thresholds converge to the collision-channel threshold [6], with MRC-SIC approaching it at a lower SNR.

For the considered setting with $\eta_0 = 10$, however, the relative performance of the two schemes for $\kappa \in 2, 3$ depends on the SNR. For $\kappa = 2$, in the low-SNR regime ($\bar{\gamma} < 12$ dB), MRC-SIC achieves higher BP thresholds owing to its combining gain, which helps raise the effective SNR above the decoding threshold η_0. In contrast, because SC-SIC does not exploit combining gain, the SINRs at all L antennas tend to remain below η_0, resulting in decoding failures.

For the considered setting with $\eta_0 = 10$, a crossover occurs near $\bar{\gamma} = 12$ dB for $\kappa = 2$, above which SC-SIC achieves a higher BP threshold. At higher SNRs, SC-SIC is more likely to find a packet–antenna pair with SINR above η_0, whereas residual multiuser interference limits the post-combining SINR of MRC-SIC. A similar crossover occurs for $\kappa = 3$.

Figure 8(b) also shows the high-SNR BP benchmarks of SC-SIC. For $\kappa = 2$, the benchmark is obtained by substituting the singleton limit in (17) and the exact two-packet error floor in (21) into the DE recursion (31). For $\kappa = 3$, the approximate three-packet error floor in (22) is additionally used. For the fixed parameters of Fig. 8(b), the resulting limiting BP thresholds are independent of $\bar{\gamma}$ and are shown as horizontal dashed lines. The finite-SNR SC-SIC curves approach these benchmarks as $\bar{\gamma} \to \infty$; thus, any visible gap between a solid curve and its dashed benchmark represents a



(a) $\kappa = 1$, $L = 3$. **(b)** $\kappa = 2, 3$, $L = 8$.

Fig. 8: BP thresholds $G^{\mathrm{BP,blk}}$ vs. average received SNR $\bar{\gamma}$ for MRC-SIC [4] and SC-SIC: (a) $\kappa = 1$ with $L = 3$, including the collision-channel threshold ($\bar{\epsilon}_{1,1} = 0$) [6]; (b) $\kappa = 2, 3$ with $L = 8$. ($\Lambda(x) = x^3$, $\eta_0 = 10$, and $\mathrm{PLR}^* = 10^{-2}$).

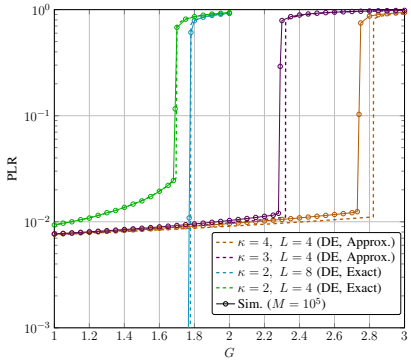


Fig. 9: PLRs vs. G of IR-SA for SC-SIC in the $R < 1$ regime, evaluated at $R \approx 0.848$ ($\eta_0 = 0.8$): $\kappa = 2, 3, 4$, $\bar{\gamma} = 0$ dB, and $\Lambda(x) = 0.760x^2 + 0.069x^3 + 0.040x^6 + 0.076x^7 + 0.033x^8 + 0.022x^{10}$.

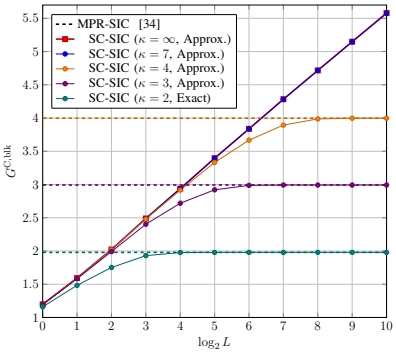


Fig. 10: Converse bounds versus $\log_2 L$ for SC-SIC with $\kappa = 2, 3, 4, 7$, and ∞, together with the corresponding κ-MPR-SIC converse bounds [34]. ($\eta_0 = 5$, $\bar{\gamma} = 20$ dB, and $d = 3$).

finite-SNR effect rather than the error floor itself.

Figure 9 validates the resulting probability expressions in the $R < 1$ regime by comparing the corresponding asymptotic DE results with simulations for $\eta_0 = 0.8$ ($R \approx 0.848$) and $\bar{\gamma} = 0$ dB for $\kappa = 2, 3$, and 4. For $\kappa = 2$, as discussed in Sec. II-B, the exact closed-form expression for the average error probability is valid over the entire range $R > 0$, and the corresponding asymptotic results show strong agreement with the simulation results for $L = 4$ and $L = 8$.

For $\kappa = 3$ and 4, the DE results based on the proposed approximation agree reasonably well with the simulations, although the discrepancies are slightly larger than in the $R \geq 1$ setting in Fig. 7. This behavior is consistent with the independence approximation in (12): conditioning on the packet selected and decoded first changes the joint SNR distribution of the residual packets, which therefore does not generally coincide with that of a fresh independent $(m-1)$-packet collision. For $R < 1$, multiple packets may satisfy the decoding condition at the same antenna during a given SIC step, and longer cancellation paths may become more important, increasing sensitivity to this approximation. Nevertheless, the approximation remains reasonably accurate for this representative $R < 1$ setting, supporting its use in evaluating the corresponding MAP thresholds and converse bounds.

From a high-SNR perspective, (20) shows that $R \approx 0.848$ satisfies the zero-floor condition for $m = 2$, but not for $m = 3$ or 4. Thus, for fixed finite L, $R < 1$ alone does not ensure a vanishing slot-level error floor for higher-order collisions.

Figure 10 illustrates the converse bound $G^{\mathrm{C,blk}}$ derived from (40) as a function of the logarithm of the number of receive antennas $\log_2 L$, alongside the corresponding κ-multi-packet reception (MPR)-SIC converse bounds derived in [34] for comparison. In κ-MPR-SIC, an AP is assumed to successfully decode all packets in a slot containing at most κ packets, i.e., $\bar{\epsilon}_{m,1} = 0$ for $m \leq \kappa$. For each finite κ, the SC-SIC converse bound monotonically converges to the corresponding κ-MPR-SIC converse bound as L increases, becoming nearly indistinguishable from the corresponding limits at $L \approx 16$, 64, and 256 for $\kappa = 2$, 3, and 4, respectively. This demonstrates that spatial diversity sufficiently reduces $\bar{\epsilon}_{m,L}$, enabling finite-antenna SC-SIC to approach the ideal MPR collision-resolution capability.

Furthermore, the two curves for $\kappa = 7$ and $\kappa = \infty$ are nearly indistinguishable throughout the plotted range $1 \leq L \leq 1024$, suggesting that, under the considered parameters, resolving collisions involving eight or more packets provides only a negligible additional increase in the converse bound.

Table IV presents the MAP thresholds $G^{\mathrm{MAP,blk}}$ and converse bounds $G^{\mathrm{C,blk}}$ for $\kappa = 2$ and 3, together with the BP thresholds $G^{\dagger\,\mathrm{BP,Conv}}$ of the spatially coupled scheme extended from the single-antenna case [7], [17]. The MAP thresholds and converse bounds increase with L, demonstrating the gain provided by spatial diversity. Moreover, the MAP thresholds closely track the corresponding converse bounds for all considered values of L and κ. The resulting converse bounds are tight because the two EXIT functions in (34) and (35) closely match each other.

In the spatially coupled scheme, a superframe consists of $L_{\mathrm{sc}} + d - 1$ frames, each containing M slots. Details of the spatially coupled construction are omitted due to space limitations and can be found in [7], [17]. The numerically obtained BP thresholds nearly coincide with the corresponding MAP thresholds. This behavior is known as the threshold saturation effect in the context of LDPC codes [35]. For both $\kappa = 2$ and $\kappa = 3$, threshold saturation is observed for all considered numbers of receive antennas.

TABLE IV
BP/MAP thresholds and converse bounds for $\kappa = 2$ and 3.
($\eta_0 = 5$, $\bar{\gamma} = 20$ dB, $\alpha = 4$, $L_{\rm sc} = 100$, $\mathrm{PLR}^* = 10^{-2}$, and $d = 3$).

	$\kappa = 2$ (Exact)			$\kappa = 3$ (Approx.)		
L	$G^{\dagger\mathrm{BP,Conv}}$	$G^{\mathrm{MAP,blk}}$	$G^{\mathrm{C,blk}}$	$G^{\dagger\mathrm{BP,Conv}}$	$G^{\mathrm{MAP,blk}}$	$G^{\mathrm{C,blk}}$
1 [7]	1.131	1.132	1.160	1.160	1.160	1.194
4	1.746	1.747	1.752	1.985	1.985	1.991
8	1.927	1.928	1.930	2.400	2.400	2.403
16	1.974	1.974	1.977	2.716	2.716	2.718
∞	1.976	1.977	1.979	2.989	2.991	2.992

VIII. Conclusion

This paper presented a slot-level analysis of multi-packet collision resolution with multi-antenna SC-SIC over Rayleigh block-fading channels. We derived exact closed-form expressions for $\bar{\epsilon}_{m,L}$ for $m \leq 2$ and developed a closed-form approximation for $m \geq 3$, both applicable to any finite L and any spectral efficiency $R > 0$. We further showed that, for $m \geq 2$ and fixed finite L, the high-SNR error floor vanishes if and only if $R \leq \log_2(m/(m-1))$, which requires $R < 1$ for $m \geq 3$. These slot-level results are directly applicable to conventional SA with L-antenna reception and enable the DE analysis of multi-antenna IR-SA. We characterized the fundamental performance limits of multi-antenna IR-SA under SC-SIC by determining the BP and MAP thresholds and deriving a converse bound. Numerical results showed that the MAP thresholds are close to the corresponding converse bounds and that spatial coupling drove the BP thresholds toward the MAP thresholds, demonstrating threshold saturation.

Future work includes exact analysis of higher-order collisions, analysis under imperfect CSI, and extensions to other slot-based RA protocols, spatially correlated antenna channels, alternative combining schemes, and other fading models.

Appendix I
Closed-Form Probability Expressions

A. $R \geq 1$

We present probability expressions for the regions shown in Fig. 3. Due to symmetry, we focus on the range where $\gamma_k > \gamma_l$. We derive the probabilities P_{A} to P_{E} as follows.

$$P_{\mathsf{A}} = \frac{e^{-\frac{1}{\bar{\gamma}}((1+\eta_0)^2-1)}}{1+\eta_0}, \quad P_{\mathsf{B}} = \frac{e^{-\frac{\eta_0}{\bar{\gamma}}}}{1+\eta_0} - \frac{e^{-\frac{1}{\bar{\gamma}}((1+\eta_0)^2-1)}}{1+\eta_0},$$

$$P_{\mathsf{C}} = \frac{1}{2}e^{-\frac{2\eta_0}{\bar{\gamma}}} - \frac{e^{-\frac{1}{\bar{\gamma}}((1+\eta_0)^2-1)}}{1+\eta_0},$$

$$P_{\mathsf{D}} = \frac{\eta_0 e^{-\frac{\eta_0}{\bar{\gamma}}}}{1+\eta_0} - e^{-\frac{2\eta_0}{\bar{\gamma}}} + \frac{e^{-\frac{\eta_0(2+\eta_0)}{\bar{\gamma}}}}{1+\eta_0}, \; P_{\mathsf{E}} = \frac{1}{2}\left(1 - e^{-\frac{\eta_0}{\bar{\gamma}}}\right)^2.$$

B. $0 < R < 1$

We present analytical probability expressions of $P_{(\mathsf{A}'\cup\mathsf{F}')}$ and $P_{\mathsf{C}'}$, whose corresponding regions are shown in Fig. 4.

To unify the two cases, as in the case of $R \geq 1$, we give the probabilities for regions $(\mathsf{A}'\cup\mathsf{F}')$ and C' ($\gamma_k > \gamma_l$):

$$P_{(\mathsf{A}'\cup\mathsf{F}')} = \frac{1}{1+\eta_0}e^{-\frac{\eta_0^2+2\eta_0}{\bar{\gamma}}} - \frac{1-\eta_0}{2(1+\eta_0)}e^{-\frac{2\eta_0}{\bar{\gamma}(1-\eta_0)}},$$

$$P_{\mathsf{C}'} = \frac{1-\eta_0}{2(1+\eta_0)}e^{-\frac{2\eta_0}{\bar{\gamma}(1-\eta_0)}} - \left(\frac{1}{1+\eta_0}e^{-\frac{\eta_0^2+2\eta_0}{\bar{\gamma}}} - \frac{1}{2}e^{-\frac{2\eta_0}{\bar{\gamma}}}\right).$$

Here, the probabilities for regions $(\breve{\mathsf{A}}' \cup \breve{\mathsf{F}}')$ and $\breve{\mathsf{C}}'$ ($\gamma_k < \gamma_l$) are omitted due to symmetry.

Appendix II
Derivation of $D(m, 1, L)$

A. Derivation of the Multi-Antenna Decoding Probability from the Single-Antenna Decoding Probability

For $R > 0$, we derive an exact expression for $D(m, 1, L)$, defined as the probability that a reference packet among m colliding packets is selected and decoded at the first SIC step with L receive antennas. Although multiple packets may simultaneously satisfy the decoding condition when $0 < R < 1$, only one packet is selected for decoding at the first SIC step.

We first relate $D(m, 1, L)$ to the corresponding single-antenna probability $D(m, 1, 1)$, defined as the probability that the reference packet is selected and decoded at the first SIC step. Since the received SNRs are independent across antennas, the probability that no packet satisfies the decoding condition at the first SIC step at any of the L antennas is $\{1 - mD(m, 1, 1)\}^L$. Consequently, the probability that at least one packet is decodable at the first SIC step across the L antennas is $1 - \{1 - mD(m, 1, 1)\}^L$. By symmetry among the m packets, we therefore obtain

$$D(m, 1, L) = \frac{1}{m}\left[1 - \{1 - mD(m, 1, 1)\}^L\right]. \tag{45}$$

B. Derivation of the Single-Antenna Decoding Probability

It remains to derive the single-antenna probability $D(m, 1, 1)$. Let $\mathcal{D}_i \triangleq \{\frac{\gamma_i}{1+\sum_{j\neq i}\gamma_j} > \eta_0\}$ denote the event that packet i satisfies the decoding condition. Whenever $\bigcup_{i=1}^m \mathcal{D}_i$ occurs, one of the packets satisfying the decoding condition is selected and decoded at the first SIC step.

Under the packet-label-symmetric selection rule of SC-SIC, the events that individual packets are selected and decoded first are mutually exclusive and equiprobable. Therefore,

$$D(m, 1, 1) = \frac{1}{m}\Pr\left(\bigcup_{i=1}^m \mathcal{D}_i\right). \tag{46}$$

To derive $D(m, 1, 1)$, we next evaluate $\Pr\left(\bigcup_{i=1}^m \mathcal{D}_i\right)$ using the inclusion–exclusion principle. Specifically, by the exchangeability of the packet SNRs,

$$\Pr\left(\bigcup_{i=1}^m \mathcal{D}_i\right) = \sum_{s=1}^m (-1)^{s+1}\binom{m}{s}\Pr\left(\bigcap_{i=1}^s \mathcal{D}_i\right). \tag{47}$$

Here, $\Pr\left(\bigcap_{i=1}^s \mathcal{D}_i\right)$, for $1 \leq s \leq m$, denotes the probability that all packets in an arbitrary subset of s packets simultaneously satisfy the decoding condition. The remainder of this appendix derives this intersection probability.

To evaluate the intersection probability for an arbitrary subset of s packets, let $X \triangleq \sum_{i=1}^s \gamma_i$ and $Y \triangleq \sum_{i=s+1}^m \gamma_i$. Since the γ_i are independent exponentially distributed RVs with mean $\bar{\gamma}$, X follows a Gamma distribution with shape parameter s. For $1 \leq s < m$, Y follows a Gamma distribution with shape parameter $m - s$, whereas for $s = m$, $Y = 0$ deterministically [30]. The corresponding PDFs are

$$f_X(x) = \frac{x^{s-1}}{(s-1)!\,\bar{\gamma}^s}e^{-\frac{x}{\bar{\gamma}}}, \; x \geq 0, \; 1 \leq s \leq m, \tag{48}$$

$$f_Y(y) = \frac{y^{m-s-1}}{(m-s-1)!\,\bar{\gamma}^{m-s}} e^{-\frac{y}{\bar{\gamma}}},\ y \geq 0,\ 1 \leq s < m. \quad (49)$$

Because X and Y are sums of disjoint sets of independent RVs, they are mutually independent, yielding $f_{X,Y}(x,y) = f_X(x) f_Y(y)$. By the law of total probability,

$$\Pr\left(\bigcap_{i=1}^{s} \mathcal{D}_i\right) = \int_0^\infty f_Y(y) \int_0^\infty \Pr\left(\bigcap_{i=1}^{s} \mathcal{D}_i \middle| X = x, Y = y\right) \times f_X(x)\, dx\, dy. \quad (50)$$

For $1 \leq s < m$, since Y is independent of $\gamma_1, \ldots, \gamma_s$, conditioning additionally on $Y = y$ does not affect the conditional joint PDF of $\gamma_1, \ldots, \gamma_{s-1}$ given $X = x$. Applying Bayes' rule and dividing the joint PDF of $\gamma_1, \ldots, \gamma_{s-1}, X$ by the marginal PDF $f_X(x)$ yields the conditional joint PDF

$$f_{\gamma_1,\ldots,\gamma_{s-1}|X}(\gamma_1, \ldots, \gamma_{s-1}|x) = \frac{\frac{1}{\bar{\gamma}^s} e^{-x/\bar{\gamma}}}{\frac{x^{s-1}}{(s-1)!\,\bar{\gamma}^s} e^{-x/\bar{\gamma}}} = \frac{(s-1)!}{x^{s-1}}, \quad (51)$$

for $\gamma_i \geq 0$ and $\sum_{i=1}^{s-1} \gamma_i \leq x$.

Conditioned on $X = x$ and $Y = y$, the decoding condition for packet $i \in \{1, \ldots, s\}$ requires $\gamma_i/(1 + x + y - \gamma_i) > \eta_0$, or equivalently,

$$\gamma_i > a(x,y), \quad \text{where } a(x,y) \triangleq \frac{\eta_0(1+x+y)}{1+\eta_0}. \quad (52)$$

Substituting $\gamma_s = x - \sum_{i=1}^{s-1} \gamma_i$ into the decoding condition for packet s and rearranging yields the upper bound constraint:

$$\sum_{i=1}^{s-1} \gamma_i < x - a(x,y). \quad (53)$$

Thus, the conditional probability is obtained by integrating the conditional joint PDF over the feasible region, i.e.,

$$\Pr\left(\bigcap_{i=1}^{s} \mathcal{D}_i \middle| X = x, Y = y\right) = \int_{\mathcal{R}_s(x,y)} \frac{(s-1)!}{x^{s-1}}\, d\gamma_1 \cdots d\gamma_{s-1}, \quad (54)$$

where the integration region $\mathcal{R}_s(x,y)$ is defined by the lower and upper bounds as

$$\mathcal{R}_s(x,y) \triangleq \left\{(\gamma_1, \ldots, \gamma_{s-1}) : \gamma_i > a(x,y),\ i = 1, \ldots, s-1,\ \sum_{i=1}^{s-1} \gamma_i < x - a(x,y)\right\}. \quad (55)$$

By shifting each integration variable by the common lower bound, i.e., introducing $u_i = \gamma_i - a(x,y)$, the integration region $\mathcal{R}_s(x,y)$ becomes $u_i > 0$, $\sum_{i=1}^{s-1} u_i < x - sa(x,y)$, which is a standard $(s-1)$-dimensional simplex. Evaluating the integral over this region yields

$$\Pr\left(\bigcap_{i=1}^{s} \mathcal{D}_i \middle| X = x, Y = y\right) = \begin{cases} \left[\frac{\{1-(s-1)\eta_0\}x - s\eta_0(1+y)}{(1+\eta_0)x}\right]^{s-1}, & x > x^\star, \\ 0, & \text{otherwise}, \end{cases} \quad (56)$$

where the threshold $x^\star \triangleq \frac{s\eta_0(1+y)}{1-(s-1)\eta_0}$ is obtained from the feasibility condition $x - sa(x,y) > 0$, provided that $1 - (s-1)\eta_0 > 0$. If this feasibility condition is violated, the feasible region is empty, and hence the selected s packets cannot simultaneously satisfy the decoding condition.

Substituting (56) into (50) gives

$$\Pr\left(\bigcap_{i=1}^{s} \mathcal{D}_i\right) = \left(\frac{1-(s-1)\eta_0}{1+\eta_0}\right)^{m-1} e^{-\frac{s\eta_0}{\bar{\gamma}\{1-(s-1)\eta_0\}}}, \quad (57)$$

for $1 - (s-1)\eta_0 > 0$, and zero otherwise.

Finally, substituting (57) into (45) yields the explicit expression for $D(m,1,L)$:

$$D(m,1,L) = \frac{1}{m}\left[1 - \left\{1 - \sum_{\substack{1 \leq s \leq m \\ 1-(s-1)\eta_0 > 0}} (-1)^{s+1} \binom{m}{s} \left[\frac{1-(s-1)\eta_0}{1+\eta_0}\right]^{m-1} \times e^{-\frac{s\eta_0}{\bar{\gamma}\{1-(s-1)\eta_0\}}}\right\}^L\right]. \quad (58)$$

For $m = 1$ and 2, (58) reduces to (6) and (7), respectively.

Appendix III
Derivation of the High-SNR Error Floor

A. Exact High-SNR Error Floor for $m = 2$

We first derive the exact high-SNR error floor for the two-packet case in (21). For $\eta_0 > 1$, the high-SNR limits of the region probabilities are $\lim_{\bar{\gamma}\to\infty} P_{\mathsf{A}} = \frac{1}{1+\eta_0}$, $\lim_{\bar{\gamma}\to\infty} P_{\mathsf{C}} = \frac{\eta_0 - 1}{2(1+\eta_0)}$, and all the other region probabilities tend to zero as $\bar{\gamma} \to \infty$. Taking the high-SNR limit of $D(2,1,L)$ in (7) and $D(2,2,L)$ in (9) yields

$$\lim_{\bar{\gamma}\to\infty} D(2,1,L) = \lim_{\bar{\gamma}\to\infty} D(2,2,L) = \frac{1}{2}\left\{1 - \left(\frac{\eta_0 - 1}{\eta_0 + 1}\right)^L\right\}.$$

Using (4), we obtain

$$\lim_{\bar{\gamma}\to\infty} \bar{\epsilon}_{2,L} = \left(\frac{\eta_0 - 1}{\eta_0 + 1}\right)^L, \qquad \eta_0 > 1.$$

B. Approximate High-SNR Error Floor for $m \geq 3$

We next derive the approximate high-SNR error floor for $m \geq 3$ over the entire range $R > 0$, or equivalently, $\eta_0 > 0$. For each integer s satisfying $1 - (s-1)\eta_0 > 0$, we have $\lim_{\bar{\gamma}\to\infty} e^{-\frac{s\eta_0}{\bar{\gamma}\{1-(s-1)\eta_0\}}} = 1$.

Therefore, taking the high-SNR limit of (58) yields

$$\lim_{\bar{\gamma}\to\infty} D(r,1,L) = \frac{1}{r}\left\{1 - \left[1 - \sum_{\substack{1 \leq s \leq r \\ 1-(s-1)\eta_0 > 0}} (-1)^{s+1} \binom{r}{s} \left(\frac{1-(s-1)\eta_0}{1+\eta_0}\right)^{r-1}\right]^L\right\}$$

$$= \frac{1}{r}\Phi_{r,L}(\eta_0), \tag{59}$$

where $\Phi_{r,L}(\eta_0)$ is defined in (23). Moreover, the exact two-packet result derived in the preceding subsection gives $\lim_{\bar{\gamma}\to\infty} D(2,L) = \Phi_{2,L}(\eta_0)$.

Taking the high-SNR limit of the closed-form approximation in (14) and substituting (59), we obtain

$$\lim_{\bar{\gamma}\to\infty} \widetilde{D}(m,L) = \frac{1}{m}\left[\sum_{j=3}^{m}\prod_{r=j}^{m}\Phi_{r,L}(\eta_0) + 2\prod_{r=2}^{m}\Phi_{r,L}(\eta_0)\right], \tag{60}$$

where we used $\frac{(m-1)!}{(j-1)!}\prod_{r=j}^{m} r^{-1} = 1/m$ and $(m-1)!\prod_{r=3}^{m} r^{-1} = 2/m$.

Finally, since

$$\bar{\epsilon}_{m,L}^{\text{floor}}(\eta_0) \approx 1 - \lim_{\bar{\gamma}\to\infty} \widetilde{D}(m,L), \tag{61}$$

(22) follows directly.

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